OntoCoDec: Type-Constrained Information Extraction via Compiled Ontology Constraints

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Abstract

We present OntoCoDec, which compiles Information Extraction (IE) ontology schemas into two composable token-level decoding constraints: (1) a *grammar layer* (XGrammar) that guarantees JSON structural validity and vocabulary membership; and (2) a *semantic layer* that enforces per-triple domain/range type constraints via a stateful entity registry and dynamic logit masking, with grammar-first fallback when constraints conflict. Without constraints, Large Language Model (LLM) based IE produces systematic domain/range (DR) violations that persist after fine-tuning; violation rates correlate with *constraint tightness* rather than ontology or model scale. Composed with Low-Rank Adaptation (LoRA), OntoCoDec achieves near-lossless composability under fine-tuning: DR $\leq 0.20\%$ with R-F1 within ± 0.6 pp. Under zero-shot decoding, DR drops by 76–99% for calibrated models, with an accompanying cross-field consistency (CF) increase (9–21% zero-shot, resolved to $\leq 1.2\%$ by fine-tuning on both RE and EE). Beam search with LoRA eliminates all four violation types on SciERC. The framework extends to event extraction without modification (preliminary, 20-document test set). OntoCoDec is training-free and applies to any autoregressive LM across four models (8B–70B) and three tokenizer families.

1 Introduction

Large language models are increasingly used for information extraction (Li et al., 2023; Sainz et al., 2024; Wei et al., 2023), but when generating relation triples (h, r, t), neither entity is guaranteed to carry a type consistent with the ontology's domain/range constraints for r. We call such outputs *domain/range violations*: structurally valid triples that are ontologically invalid. Violations persist across the capability spectrum: 15.4% of triples on SciERC (Luan et al., 2018) and 18.5% on DocRED (Yao et al., 2019) under zero-shot decoding, and 1–2% even after LoRA fine-tuning; rates correlate with *constraint tightness* (few valid type pairs per relation) rather than ontology or model scale. Type violations are ontologically invalid regardless of downstream application: a triple whose head or tail type lies outside the relation's domain/range specification contradicts the schema definition. At deployment-grade quality (R-F1 40–46%), where extracted triples feed directly into knowledge graphs, reliable type-safety enforcement is needed.

Existing approaches leave a fundamental gap. Schema-informed prompting reduces violations by 62% on average but is per-relation unreliable (violations for EVALUATE-FOR actually *increase* by 5.8 pp). Grammar-constrained decoding (Dong et al., 2025; Willard and Louf, 2023; Beurer-Kellner et al., 2023) guarantees syntactic well-formedness but cannot encode cross-field semantic constraints. Post-hoc filtering achieves higher F1 in most zero-shot settings via precision gains, but at the cost of discarding up to 48% of predictions including true positives; at high rejection rates (e.g., Mistral DocRED 73.7%), it discards the majority of outputs and becomes impractical. OntoCoDec preserves recall by enforcing constraints during generation rather than discarding non-compliant outputs. Training-based approaches (Sainz et al., 2024; Lu et al., 2022) provide only soft guidance that cannot eliminate violations.

We propose **OntoCoDec** (**Onto**logy-**Co**mpiled **Dec**oding), a framework that compiles an IE ontology into two composable layers of decoding-time constraints. Our two-layer architecture reflects the observation that structural violations (malformed output, unknown types) and semantic violations (type-constraint breaches) require fundamentally different enforcement mechanisms. The *grammar layer* enforces JSON well-formedness and vocabulary validity via XGrammar. The *se-*

mantic layer maintains a stateful entity registry during generation: as entities are decoded, the registry records type assignments; when relation arguments are generated, dynamic logit masking restricts output to types compatible with the relation's domain/range specification. The layers compose via bitwise AND with an *enabling dependency*: the grammar layer guarantees vocabulary membership, so the semantic layer can look up domain/range rules. Unlike type-constrained code generation where types are statically declared, IE requires a dynamic entity registry built during generation, with constraint targets that depend on the co-occurring relation. Composed with LoRA, OntoCoDec achieves near-lossless composability under fine-tuning: DR $\leq 0.20\%$ at $<$0.6 pp F1 cost (R-F1 40–46), our recommended deployment configuration. Zero-shot evaluation, which stress-tests constraint robustness without task supervision (R-F1 2–8%), confirms DR drops by 76–99% for calibrated models while remaining F1-neutral.

Our contributions:

1. A systematic study of ontological constraint violations in LLM-based IE: rates correlate with constraint tightness rather than ontology scale, and prompt-level intervention is per-relation unreliable.

2. OntoCoDec, a two-layer framework compiling ontology schemas into stateful logit masks that enforce pair-based domain/range constraints during autoregressive IE, unlike code generation where types are statically declared.

3. Near-lossless composability with LoRA on RE: DR $\leq 0.20\%$ at $<$0.6 pp F1 cost (R-F1 40–46), with CF trade-off resolved to $\leq$1.2% by fine-tuning. Zero-shot evaluation across four models (8B–70B) confirms DR reductions of 76–99% for calibrated models.

4. Beam search (k=4) with LoRA eliminates all four violation types on SciERC. Preliminary extension to event extraction (WikiEvents, 20-document test set): vocabulary violations eliminated without architectural modification; fine-tuning composability analysis.

2 Background

IE Ontology Structure. An IE ontology $\mathcal{O} = (\mathcal{E}, \mathcal{R}, \sigma)$ defines entity types $\mathcal{E}$, relation types $\mathcal{R}$, and a mapping $\sigma : \mathcal{R} \to \mathcal{P}(\mathcal{E} \times \mathcal{E})$ specifying valid (head type, tail type) pairs per relation. A triple (h, r, t) is *type-valid* if $(\text{type}(h), \text{type}(t)) \in \sigma(r)$. We quantify constraint restrictiveness via *constraint tightness* $\kappa(r)$ (Eq. 3, §4): tighter constraints leave fewer valid type pairs per relation. We define ontology constraints broadly to encompass both formal ontology specifications (e.g., the KAIROS event ontology with explicit role-entity type mappings) and empirically derived type-pair constraints extracted from annotated training data. The latter applies when datasets lack a formal ontology but provide typed annotations (SciERC, DocRED); OntoCoDec compiles both sources into the same constraint representation without modification. Empirical constraints achieve 99%+ instance-level coverage on held-out data (SciERC 99.0%, DocRED 99.4%); uncovered type pairs trigger grammar-first fallback (§4), ensuring no valid triples are blocked.

Grammar-Constrained Decoding. Grammar-constrained decoding restricts each token y_t to a mask $M_g \in \{0,1\}^{|V|}$ derived from a target grammar. XGrammar (Dong et al., 2025) compiles JSON Schemas into pushdown automata for efficient enforcement of structural constraints; Outlines (Willard and Louf, 2023) and SGLang (Zheng et al., 2024) offer similar capabilities. However, these methods are limited to context-free constraints and cannot enforce cross-field semantic rules (e.g., "if r = FEATURE-OF, then $\text{type}(h) \in \{\text{METHOD}, \text{OTHERSCIENTIFICTERM}\}$"), which depend on previously generated field values.

Problem Statement. Given $\mathcal{O}$ and an autoregressive LM $\mathcal{M}$, we seek constraints $\mathcal{C}$ ensuring: (a) **structural validity**, i.e., well-formed JSON with $\mathcal{E}$/$\mathcal{R}$ membership; and (b) **type validity**, i.e., $(\text{type}(h), \text{type}(t)) \in \sigma(r)$ for every triple. Condition (a) is context-free; (b) is context-sensitive but *prefix-decidable*: at the point where a type token is generated, all information needed (relation type, entity registry) is available in the decoded prefix. This requires that each entity type name maps to a distinguishable token sequence (§3.3).

3 Method

We present OntoCoDec, a two-layer constrained decoding framework that compiles ontology schemas into token-level masks enforced during autoregressive generation. Unlike type-constrained code generation, IE presents three distinct challenges:

(1) entity types are inferred from free text rather than declared annotations, (2) the entity registry is built dynamically during generation, and (3) constraint targets are context-dependent type assignments rather than fixed grammar productions. The grammar layer (§3.2) guarantees structural validity (well-formed JSON with ontology-vocabulary membership), while the semantic layer (§3.3) enforces domain/range type constraints via a stateful entity registry. We describe the problem formulation (§3.1), each layer, and two decoding strategies (§3.4). Formal analysis of constraint composition is deferred to Section 4.

3.1 Problem Formulation

IE as structured generation. Given an input text x, a language model generates a structured JSON output y:

$$y = \{\texttt{entities} : [\{n_i, \tau_i\}],\ \texttt{relations} : [\{h_j, t_j, r_j, \tau_j^h, \tau_j^t\}]\} \quad (1)$$

where each entity has a name n_i and type $\tau_i \in \mathcal{E}$, and each relation specifies head h_j, tail t_j, relation type $r_j \in \mathcal{R}$, and head/tail entity types τ_j^h, τ_j^t. The output JSON Schema (§3.2) enforces a fixed field ordering (head $\rightarrow$ tail $\rightarrow$ relation $\rightarrow$ head_type $\rightarrow$ tail_type) via deterministic grammar compilation. This ordering ensures that the relation type r_j is known before entity type fields are generated, which enables domain and range lookups in sequential masking (§3.3).

Ontology compliance. Using the ontology $\mathcal{O} = (\mathcal{E}, \mathcal{R}, \sigma)$ defined in §2, we write $\mathrm{dom}(r) = \{\tau^h : \exists \tau^t,\ (\tau^h, \tau^t) \in \sigma(r)\}$ for domain-valid head types and $\mathrm{range}(r, \tau^h) = \{\tau^t : (\tau^h, \tau^t) \in \sigma(r)\}$ for the conditional range given head type τ^h. Our enforcement target is *pair-based* type validity: $(\tau^h, \tau^t) \in \sigma(r)$ for every triple. The decoder enforces this sequentially: $\tau^h \in \mathrm{dom}(r)$ at head position, then $\tau^t \in \mathrm{range}(r, \tau^h)$ at tail position. This is equivalent to the full pair check and strictly stronger than independent domain/range enforcement, because most relations have non-rectangular type signatures ($\sigma(r) \subsetneq \mathrm{dom}(r) \times \mathrm{range}(r)$ for all 7 SciERC and 73/96 DocRED relations; see §4).[1] Our goal is maximizing $P(y \mid x)$ subject to all triples being type-valid and y being well-formed JSON conforming to Eq. (1).

[1] An independent check ($\tau^h \in \mathrm{dom}(r) \wedge \tau^t \in \mathrm{range}(r)$) would over-permit type pairs by $1.50\times$ on SciERC and $1.58\times$ on DocRED, corresponding to 33.3% and 36.8% spurious type pairs.

3.2 Grammar Layer

The grammar layer compiles $\mathcal{O}$ into a CFG enforced at every decoding step via logit masking, in two stages.

Schema-to-grammar compilation. We generate a JSON Schema encoding the structure in Eq. (1) with `enum` constraints restricting entity types to $\mathcal{E}$ and relation types to $\mathcal{R}$. This schema is compiled into an EBNF grammar via XGrammar (Dong et al., 2025); the result is a pushdown automaton that tracks JSON structure and enum membership.

Logit masking. At step t, the automaton computes grammar-valid tokens $G(t) \subseteq V$; all others receive logits $-\infty$. This simultaneously enforces: (i) structural validity (well-formed JSON), (ii) relation vocabulary membership ($r_j \in \mathcal{R}$), and (iii) entity type membership ($\tau_i \in \mathcal{E}$). Domain/range constraints are *not* enforced, as they depend on entity types within each triple; this gap is filled by the semantic layer.

3.3 Semantic Layer

The semantic layer enforces domain/range constraints through a stateful logits processor maintaining an entity registry during generation.

Entity registry. The semantic layer builds a registry $\mathcal{D}$: name $\rightarrow$ type incrementally during entity generation: each completed entry $\{n_i, \tau_i\}$ adds (n_i, τ_i) to $\mathcal{D}$, which is consulted during relation generation for type lookups. At each step t, the processor pattern-matches the decoded prefix to identify the current field position; semantic masks activate only at type-bearing positions (HEADTYPE, TAILTYPE).

Dynamic masking. At type-bearing positions, the processor computes a semantic mask $S(t) \subseteq V$ restricting generation to type-valid tokens.

Head type masking. At a HEADTYPE position, the processor considers the entity registry $\mathcal{D}$ and the current relation type r_j (if already generated). If $h_j \in \mathcal{D}$ and $\mathcal{D}[h_j] \in \mathrm{dom}(r_j)$, the mask allows only that registered type: $S(t) = \{\mathrm{tok}(\mathcal{D}[h_j])\}$, ensuring consistency with both the entity list and domain constraints. Otherwise, if r_j is known, the mask falls back to all domain-valid types: $S(t) = \{\mathrm{tok}(\tau) : \tau \in \mathrm{dom}(r_j)\}$. If r_j is unavailable, no semantic mask is applied.

Tail type masking. At a TAILTYPE position, the head type τ_j^h is already determined from the decoded prefix. The processor computes $\text{range}(r_j, \tau_j^h)$, the valid tail types conditioned on the generated head type (§3.1). If $t_j \in \mathcal{D}$, the mask further restricts to the registered type when it belongs to this set. This sequential check ($\tau^h \in \text{dom}(r)$, then $\tau^t \in \text{range}(r, \tau^h)$) achieves exact pair-based enforcement, strictly tighter than an independent range check $\tau^t \in \text{range}(r)$.

Tokenizer-agnostic type masking. Mapping type names to token IDs is non-trivial: SentencePiece tokenizers produce context-dependent tokenizations, and prefix-shared types (e.g., SEN, SID, SIDE in WikiEvents) share first tokens. We use *continuation masking*: at compile time, a prefix map identifies all tokens whose decoded text is a valid prefix of each type name; at decode time, the mask allows any continuation token, disambiguating over multiple steps. This handles byte-level BPE (Qwen, Llama) and SentencePiece (Mistral) without tokenizer-specific logic.

Grammar-semantic interaction. The final mask is $G(t) \cap S(t)$; when empty, grammar-only control applies (*grammar-first fallback*). This embodies an *enabling dependency*: semantic-only constraints reduce DR by only 24% without grammar-enforced vocabulary (§4).

3.4 Decoding Strategies

S1: Single-pass decoding. Both constraint layers are active throughout a single generation pass; the entity registry is populated during entity generation and immediately available for relation-level type checking.

S2: Two-pass decoding. A first pass extracts entities under grammar constraints; extracted entities compile an input-dependent grammar for the second pass, which generates relations under both grammar and semantic constraints. S1 achieves higher R-F1 at comparable DR (§5.1) and is the primary configuration.

4 Constraint Composition

At each decoding step t, let $G(t) \subseteq V$ and $S(t) \subseteq V$ denote the tokens permitted by the grammar and semantic layers, respectively. The final mask is their conjunction:

$$M_{\text{final}}(t) = M_{\text{grammar}}(t) \ \wedge \ M_{\text{semantic}}(t), \quad (2)$$

so a token is allowed only if it satisfies *both* structural and type-level constraints. The conjunction is asymmetric: $S(t)$ is defined only over $G(t)$. This reflects the enabling dependency (§3): the semantic layer requires the relation type to belong to $\mathcal{R}$, a property guaranteed only by the grammar layer.

Grammar-First Fallback. When $S(t) \cap G(t) = \emptyset$ (tight constraints leave no token that is both grammar-valid and semantically valid), naïve masking would force selection of a grammar-invalid token, breaking JSON structure. We enforce a *grammar-first fallback*: if no semantically valid token survives grammar filtering, semantic masking is skipped and the grammar layer alone controls selection. Structural integrity is a hard constraint; a type violation is a soft error detectable by downstream validation. Among type violations, DR and CF differ in severity: a DR violation renders the triple ontologically invalid (the entity type lies outside the relation's domain or range), whereas a CF violation means each triple is individually schema-valid but the same entity receives different type assignments across triples.

Constraint Tightness. Recall the tightness of relation r (§2):

$$\kappa(r) = 1 - \frac{|\sigma(r)|}{|\mathcal{E}|^2}, \quad (3)$$

The mean tightness $\bar{\kappa} = \frac{1}{|\mathcal{R}|} \sum_r \kappa(r)$ characterizes ontology restrictiveness. Tighter constraints eliminate more grammar-valid tokens at type-bearing positions; fallback frequency and residual DR increase accordingly. This dual-factor pattern (tightness $\times$ baseline violation rate) is consistent with the cross-dataset results: SciERC ($\bar{\kappa}$=0.333) achieves 94–99% DR reduction vs. DocRED ($\bar{\kappa}$=0.842) at 45–90%; the full range across three models is 42–99% (§5.3; Appendix C). The pattern extends to WikiEvents EE ($\bar{\kappa}$=0.82; Appendix C).

Non-Rectangularity and Sequential Masking. Sequential masking is necessary because ontology type signatures are non-rectangular (0/7 SciERC, 23/96 DocRED relations satisfy $\sigma(r) = \text{dom}(r) \times \text{range}(r)$); independent enforcement would admit 33–37% spurious pairs.

Composability with fine-tuning. Fine-tuning narrows the fallback regime: DR drops to $\leq 0.20\%$ and CF to $\leq 1.18\%$, with R-F1 within ± 0.6 pp; the

grammar layer becomes a no-op and the semantic layer alone reduces residual DR by 39–72% (§5.1).

5 Experiments

We evaluate OntoCoDec in two complementary regimes. Fine-tuning composability (§5.1) demonstrates deployment readiness: constraints compose losslessly with standard LoRA training at R-F1 40–46, our recommended configuration. Zero-shot evaluation stress-tests constraint robustness under adversarial conditions (no task supervision, R-F1 2–8), where CF trade-offs are expected and documented (§5.3).

Setup. We evaluate on SciERC (Luan et al., 2018) (551 test sentences, 7 relations, $\bar{\kappa}$=0.333) and DocRED (Yao et al., 2019) (998 dev documents,[2] 96 relations, $\bar{\kappa}$=0.842). RE type constraints are extracted automatically from observed type pairs in training data (neither dataset provides a formal ontology); EE constraints on WikiEvents (§5.4) are derived from the KAIROS ontology. The framework handles both sources without modification. Primary model: Qwen3-8B (Qwen Team, 2025), zero-shot greedy decoding; cross-model validation in §5.3, composability with fine-tuning in §5.1. We measure four violation categories: **DR** (domain/range type pair $\notin \sigma(r)$), **RV** (relation $\notin \mathcal{R}$), **ET** (entity type $\notin \mathcal{E}$), and **CF** (same entity assigned conflicting types across triples). Baseline DR violations are 15.37% (SciERC) and 18.46% (DocRED), comparable rates associated with constraint tightness rather than ontology scale (§1); schema-informed prompting reduces DR by 62% on SciERC but is per-relation unreliable (Appendix C).

5.1 Ablation and Composability with Fine-Tuning

Table 1 presents the five-way ablation on SciERC (Qwen3-8B, zero-shot, 551 documents); only the active constraint layers differ across rows.

Zero-shot diagnostic. Full S1 reduces DR from 15.37% to 0.93% (↓94%) while R-F1 is unchanged (8.23 → 8.36, p=0.58). Per-relation DR drops to ≤2.82%, reaching 0% on 4/7 relations (Appendix C). Post-hoc filtering achieves slightly higher F1 via precision gains but discards up to 48% of predictions including true positives (Appendix C). CF rises from 0.08% to 9.90% (§7).

[2]DocRED test annotations are not publicly available; we evaluate on the development set.

	Violation Rates (%)						F1	
Mode	DR↓	ΔDR	RV↓	ET↓	CF	Tot.↓	Rel.↑	Ent.↑
Unconstrained	15.37	—	1.39	0.40	0.08	15.45	8.23	31.01
Grammar	14.84	↓3%	**0**	**0**	0.08	14.92	8.23	30.98
Semantic	11.65	↓24%	1.51	0.40	2.71	14.36	8.15	31.01
Full (S1)	**0.93**	**↓94%**	**0**	**0**	9.90	**10.83**	**8.36**	30.98
Full (S2)	0.97	↓94%	**0**	**0**	8.03	**9.00**	8.03	**31.90**

Table 1: Five-way ablation on SciERC (551 docs, Qwen3-8B, zero-shot). ΔDR: relative change vs. unconstrained. Tot. = DR + CF (total type-violation burden). Best per column in **bold** (CF excluded; Tot. included). F1 ×100. Bootstrap: 10,000 iterations. Grammar-only represents the ceiling of grammar-constrained decoding (Geng et al., 2023): vocabulary membership is enforced but cross-field semantic constraints are not.

Layer contributions. Grammar-only eliminates RV and ET with no F1 impact while DR remains at 14.84%, the ceiling of grammar-based decoding (Geng et al., 2023); grammar-only CF is near-zero (0.08%), so CF originates from the semantic layer. Semantic-only reduces DR by only 24% and cannot eliminate RV or ET; this is the *enabling dependency*: adding the grammar layer yields a further 92% reduction (11.65% → 0.93%).

S1 vs. S2. S2 achieves comparable DR (0.97%) but lower R-F1 (8.03 vs. 8.36) with fewer triples (1,644 vs. 1,729) and does not resolve the CF trade-off (8.03% vs. 9.90%); we adopt S1 as the primary configuration.

Composability with fine-tuning. Fine-tuning alone reduces DR to 1–2% but leaves residual violations (Table 2). Adding S1: Qwen DR 1.95% → 0%, Llama 1.11% → 0.09% on SciERC; Qwen 0.76% → 0.01%, Llama 0.49% → 0.20% on DocRED. Across all settings, DR ≤ 0.20% with R-F1 within ±0.6 pp. CF interference drops from +9.82 pp (zero-shot) to +1.18 pp (fine-tuned), tracing to GENERIC↔X ambiguity (Appendix C). Beam search (k=4) eliminates residual DR and CF to <1% (Appendix F).

5.2 Cross-Dataset Validation on DocRED

Table 3 presents the four-way ablation on DocRED (96 Wikidata relation types, 14× SciERC; Qwen3-8B, zero-shot, 998 documents).

Zero-shot diagnostic. Full S1 reduces DR from 18.46% to 10.21% (↓45%), eliminates ET (→ 0%),

Model	Setting	Mode	DR ↓	CF	R-F1 ↑	E-F1 ↑	ΔDR
Qwen3-8B	Zero-shot	Unconstr.	15.37	0.08	8.23	31.01	—
		Full S1	0.93	9.90	8.36	30.98	↓94%
	LoRA	Unconstr.	1.95	0	43.86	67.05	—
		Full S1	**0**	1.18	43.77	67.05	↓100%
Llama-3.1-8B	Zero-shot	Unconstr.	13.34	0.24	2.68	23.56	—
		Full S1	0.08	6.56	2.54	23.49	↓99%
	LoRA	Unconstr.	1.11	0	45.45	67.76	—
		Full S1	0.09	0.46	45.47	67.76	↓92%

Table 2: Composability on SciERC (551 docs): zero-shot vs. LoRA × unconstrained vs. S1. Adding S1 to LoRA: DR $\leq 0.09\%$, R-F1 within ± 0.1 pp. DocRED: Qwen DR -99%, Llama DR -59%. F1 $\times 100$.

	Violation Rates (%)						F1	
Mode	DR ↓	ΔDR	RV ↓	ET ↓	CF	Tot. ↓	Rel. ↑	Ent. ↑
Unconstrained	18.46	—	29.66	2.48	0.10	18.56	7.74	58.15
Grammar	26.17	↑42%	**0.03**	**0**	0.07	26.24	**8.31**	**59.38**
Semantic	13.84	↓25%	29.87	4.22	1.05	14.89	7.70	57.81
Full (S1)	**10.21**	**↓45%**	**0.03**	**0**	2.57	**12.78**	8.23	59.37

Table 3: Four-way ablation on DocRED (998 documents, Qwen3-8B, zero-shot). ΔDR: relative change vs. unconstrained. Tot. = DR + CF (total type-violation burden). Best per column in **bold** (CF excluded; Tot. included). F1 $\times 100$. Bootstrap significance: R-F1 $+0.50$ pp ($p < 0.001$).

and near-eliminates RV (29.66% → 0.03%).[3] R-F1 improves from 7.74 to 8.23 (+0.50 pp; $p < 0.001$), attributable to elimination of 5,189 OOV relation predictions. CF increases modestly (0.10% → 2.57%), consistent with the CF↔DR trade-off (§7).

Layer contributions. Grammar-only eliminates RV and ET but *increases* DR to 26.17%: OOV triples (29.66% baseline) become evaluable and expose latent type incompatibilities. Semantic-only reduces DR by 25% but cannot eliminate RV or ET, the same enabling dependency observed on SciERC. Residual DR (10.21% vs. SciERC's 0.93%) reflects higher tightness ($\bar{\kappa}$=0.842 vs. 0.333), which forces frequent fallback; effective semantic coverage is ∼90% (Appendix C). Under fine-tuning, residual narrows to 0.01–0.20% (Table 2).

5.3 Cross-Model Validation

Table 4 replicates S1 with Llama-3.1-8B and Mistral-7B under identical prompts.

[3]The residual 0.03% RV traces to a rare XGrammar automaton edge case where subword tokens from different enum entries combine into chimeric values (e.g., `member` + `icipant_of` from `member_of` and `participant_of`). This is not a fundamental limitation of the approach.

		Violation Rates (%)				R-F1
Dataset	Model	DR ↓	RV ↓	ET ↓	CF	↑
SciERC	Qwen (U)	15.37	1.39	0.40	0.08	8.23
	Qwen (S1)	0.93	0	0	9.90	8.36
	Llama (U)	13.34	0.55	0	0.24	2.68
	Llama (S1)	0.08	0	0	6.56	2.54
	Mistral (U)	56.94	0	1.50	56.87†	3.66
	Mistral (S1)	13.40	0	0	1.27	3.67
DocRED	Qwen (U)	18.46	29.66	2.48	0.10	7.74
	Qwen (S1)	10.21	0.03	0	2.57	8.23
	Llama (U)	26.07	16.03	0.13	0.17	5.98
	Llama (S1)	2.55	0	0	12.02	6.27
	Mistral (U)	57.42	42.35	7.58	24.78	4.34
	Mistral (S1)	33.20	0	0	0.19	5.52

Table 4: Three-model zero-shot validation. U: unconstrained; S1: full two-layer. Parse failures: Llama DocRED 9.1%; others ≤1.1%. †Mistral: poor zero-shot calibration causes pervasive type inconsistencies. F1 $\times 100$.

Results. RV and ET reach ≤0.03% for all models on both datasets. DR drops 76–99% on SciERC and 42–90% on DocRED; R-F1 improves in 5/6 settings (DocRED: Qwen +0.50 pp, $p<0.001$; Llama +0.29 pp, $p=0.013$). CF direction depends on calibration: S1 introduces CF for calibrated models (up to 12.02%; §7) but reduces it for Mistral (56.87% → 1.27%). Fine-tuning resolves both: DR $\leq 0.20\%$, CF $\leq 1.18\%$, R-F1 ± 0.6 pp (§5.1).

Mistral as boundary condition. Effective semantic constraint enforcement presupposes that the base model assigns entity types within the ontology vocabulary. When entity typing quality degrades substantially (Mistral: ET 1.5–7.6% unconstrained, DR 56.9–57.4%), the semantic layer cannot correct entity misclassification errors that precede constraint application; DR reduction is limited to 42% vs. 76–99% for calibrated models.

Post-hoc filtering comparison. Post-hoc filtering achieves higher F1 in 5/6 zero-shot settings through precision gains (Appendix C), but at rejection rates of 15–74%, discarding substantial correct predictions. Under fine-tuning, both methods achieve comparable F1. OntoCoDec and post-hoc filtering target different operating points: post-hoc suits precision-critical applications willing to sacrifice recall; OntoCoDec preserves recall by enforcing constraints during generation.

Model	Mode	Violation Rates (%)				F1	
		EV ↓	RoV ↓	AC	CF	Trig. ↑	Ent. ↑
Qwen3-8B	Unconstr.	24.0	74.9	0.3	0	11.8	0
	Grammar	**0**	76.1	8.2	—	—	—
	Full S1	**0**	**0**	2.1	21.1	13.4	14.1
Llama 3.1-8B	Unconstr.	10.5	88.8	0.1	0	5.7	0
	Grammar	**0**	77.0	8.8	—	—	—
	Full S1	**0**	**0**	3.8	17.2	7.5	17.6
Mistral 7B	Unconstr.	37.3	63.9	0	0	2.4	0
	Grammar	**0**	64.0	13.8	—	—	—
	Full S1	**0**	**0**	1.3	14.9	3.9	11.7

Table 5: WikiEvents EE results (246 documents, zero-shot). Violation rates over parsed documents. —: not applicable (grammar-only has no semantic-layer metrics). F1 × 100.

5.4 Cross-Task Validation: Event Extraction

We apply OntoCoDec to event extraction (EE) on WikiEvents (Li et al., 2021) (56 event types, 80 roles, 27 entity types, 245 constraints; Appendix C) without modification. Violations extend to EV (event vocabulary), RoV (role vocabulary), and AC (argument constraint, analogous to DR).

Results. Unconstrained EE produces extreme vocabulary violations (EV+RoV >85%; Table 5), with entity F1 = 0 because the parser cannot process off-schema output. Full S1 eliminates EV and RoV across all models, which validates the tokenizer-agnostic design across byte-level BPE and SentencePiece. Grammar-only eliminates EV but leaves RoV at 64–77% and surfaces latent AC (8–14%), consistent with DocRED's pattern (§5.2); full S1 drives RoV to 0%. Residual AC (1.3–3.8%) traces to semantically similar types (e.g., SID vs. SEN). CF rises to 15–21%, exceeding RE levels (10–12%) because EE's multi-argument structures increase entity reuse across slots. Post-hoc filtering is impractical (<0.5% arguments retained). Unlike RE, EE shows measurable F1 gains: entity F1 0% → 12–18%, trigger F1 +1.5–1.8 pp.

Composability with fine-tuning (preliminary). WikiEvents uses a 194/20/20 train/dev/test split (Li et al., 2021); the 20-document test set is a dataset constraint, not a design choice. LoRA + S1 eliminates RoV (Qwen 38.31% → 0%, Llama 8.8% → 0%; bootstrap 95% CI [0%, 0%] and [0%, 1.73%]) and reduces AC to ≤0.8%, with argument F1 within ±1.1 pp; trigger F1 shows larger shifts (−5.4/−0.6 pp) with wide confidence intervals at this sample size. CF remains low under fine-tuning (0.77–1.18%), consistent with RE composability results (§4, Table 2). Constrained decoding does not further reduce EE CF because the semantic layer enforces per-event role-type compatibility but not cross-event entity type consensus, the same architectural limitation identified in RE (§4).

Runtime overhead. S1 overhead is +13.1% on SciERC and +2.43% on DocRED (amortized over longer sequences); under LoRA, +10.0% with 37% lower absolute latency. Grammar compile: one-time 44 ms (Appendix C). Scaling benchmarks confirm constant overhead (2–3%) up to 200 entity types and 1,000 relations (Appendix K).

Beam search compatibility. Under greedy S1, the grammar-first fallback resolves DR↔CF conflicts one token at a time; beam search (k=4) explores paths that satisfy both layers jointly and reduces DR and CF simultaneously. With LoRA, beam S1 eliminates all four violation types on SciERC for both Qwen and Llama (R-F1 44.23, 46.11; Table 15). Under zero-shot, beam reduces DR to 0% for calibrated models (Qwen, Llama) and CF substantially (9.90% → 0.53% Qwen; 6.56% → 0.93% Llama); Mistral remains a boundary condition (DR 7.99%, CF 1.27%) due to high baseline violations (Appendix F). Total latency overhead is 1.54× (beam 1.32× + constraint 1.17×; Table 17).

Cross-scale generalizability. On Llama-3.1-70B (AWQ INT4), OntoCoDec achieves DR→0% on SciERC (R-F1 −0.03 pp) and **0%** across ET, RV, and DR on the full DocRED 998-document set (R-F1 +0.04 pp), the only configuration to completely eliminate DR on DocRED. CF rises to 7.01% (from 0.07%); the low baseline reflects under-measurement (malformed outputs bypass the checker), while constrained decoding exposes more triples to the consistency check (§7). Model scale reduces but does not eliminate violations.

6 Related Work

We provide a detailed survey in Appendix B; here we position OntoCoDec along three axes.

Grammar-constrained decoding. XGrammar (Dong et al., 2025), Outlines (Willard and Louf, 2023), LMQL (Beurer-Kellner et al., 2023), SynCode (Ugare et al., 2024), and Grammar-Aligned Decoding (Park et al., 2024) guarantee format validity via CFG-based token

masking; JSONSchemaBench (Geng et al., 2025) benchmarks these systems. GCD (Geng et al., 2023) applies grammar constraints to IE tasks. In code generation, PICARD (Scholak et al., 2021) and Synchromesh (Poesia et al., 2022) enforce type constraints via incremental parsing, and Mündler et al. (2025) compile full type systems into prefix automata. These methods ensure syntactic or type-level validity but operate on static type declarations; IE requires a dynamic entity registry built during generation, where constraint targets depend on the co-occurring relation type. OntoCoDec's grammar layer builds on XGrammar; its semantic layer extends constrained decoding from syntax to ontology-level type enforcement.

Schema-aware IE. GoLLIE (Sainz et al., 2024) and KnowCoder (Li et al., 2024) encode ontology structure as Python dataclasses and train models to generate typed code. InstructUIE (Wang et al., 2023), DeepStruct (Wang et al., 2022), UIE (Lu et al., 2022), and USM (Lou et al., 2023) unify extraction tasks via structured output formats with task-specific training. UniversalNER (Zhou et al., 2024) achieves strong zero-shot NER via targeted distillation. Wadhwa et al. (2023) benchmark LLM-based RE and find persistent schema compliance gaps. These methods internalize constraints through training; OntoCoDec enforces them at decoding time without retraining, and the two approaches are complementary (a GoLLIE-trained model could use OntoCoDec to eliminate residual violations).

Constraint enforcement in IE. Roth and Yih (2004) first formulated hard type constraints for IE via integer linear programming; OntoCoDec achieves analogous enforcement within autoregressive decoding. Post-hoc validation (Khorshidi et al., 2025; Caufield et al., 2024; Cotti et al., 2025; Shen et al., 2025) detects violations after generation and requires rejection or correction cycles. EnriCo (Zaratiana et al., 2024) enforces structural constraints via ASP but requires hand-crafted rules; IterGen (Ugare et al., 2025) supports semantic constraints via backtracking but requires manual implementation per constraint. BoostCD (Sakota et al., 2025) combines constrained and unconstrained decoding for IE. Recent work shows constrained decoding can degrade reasoning (Banerjee et al., 2025; Schall and de Melo, 2025; Lee et al., 2026); however, extraction tasks are robust to format constraints (Tam et al., 2024), and OntoCoDec constrains only type tokens while leaving entity spans unconstrained. OntoCoDec prevents violations during decoding, is training-free, and compiles constraints automatically from ontology schemas.

7 Discussion and Conclusion

OntoCoDec compiles IE ontology schemas into composable token-level decoding constraints. The core finding is that ontology type constraints can be compiled into token-level masks that compose near-losslessly with standard fine-tuning: DR reaches $\leq 0.20\%$ with R-F1 within ± 0.6 pp on RE (R-F1 40–46), RoV$\rightarrow$0% and AC$\leq$0.8% on EE, and beam search (k=4) eliminates all four RE violation types for calibrated models (§5.4). Under zero-shot decoding (R-F1 2–8%), DR drops by 42–99% across three models and three datasets with no significant F1 cost.

CF remains an open challenge for per-slot enforcement: the current architecture enforces type constraints independently per relation slot, so the same entity can receive different type assignments across triples. Global entity-level type consensus during decoding, where all slots referencing the same entity must agree on its type, offers the most direct resolution path.

The framework applies to any autoregressive IE system without model modification or additional training: the constraint compiler takes an ontology schema as input and produces token-level masks compatible with any LM vocabulary. This generality is validated across four models (8B–70B), three tokenizer families, and two IE tasks (RE, EE). The interaction between constraint tightness and model calibration is consistent across three datasets but would benefit from validation on a broader range of ontologies and model scales.

Limitations

Constraint scope. CF impact is model-dependent: for well-calibrated models (Qwen, Llama), S1 introduces CF violations (up to 12.02%) as a consequence of per-relation independent enforcement; for poorly-calibrated models (Mistral), S1 substantially reduces CF (56.87% $\rightarrow$ 1.27%) by constraining type assignments to schema-valid values. CF fallback ablation shows that eliminating CF re-routes violations to DR (+13.85 pp). CF$\leftrightarrow$DR thus represents a fundamental trade-off in composable constraint systems rather than an independently resolvable limitation (Appendix C). Global entity-level type consensus during decoding offers the most direct path to resolving CF violations. CF type inconsistencies may propagate to downstream tasks that consume extracted triples (e.g., knowledge graph construction, question answering); the impact of such propagation has not been evaluated. Entity F1 impact is model-dependent (+1.22 pp for Qwen vs. −0.64 pp for Llama on DocRED), partially confounded by increased parse failures under constraints (Llama: 9.1% vs. 6.8%). Residual DR under fine-tuning (36 violations across all fine-tuned settings, both models and both datasets) decomposes into entity type errors (39%), schema coverage gaps (33%), and model generation errors correctly caught by constraints (28%); Llama DocRED alone accounts for 33 of 36 residual violations (92%), reflecting weaker entity typing (Appendix C). Schema coverage gaps (one-third of residual violations) arise because RE constraints are extracted from training-data type pairs, which may not cover all valid combinations; larger training sets or manual constraint table completion can reduce this gap.

Evaluation coverage. Zero-shot results cover three 7–8B-class models, where relation F1 remains around 4–8% compared to ∼45% for fine-tuned systems. Zero-shot evaluation covers three 8B-class and one 70B-class model; fine-tuning composability is validated only on 8B-class models. Scaling to smaller models has not been verified, and the dual-factor pattern's predictive scope may differ at other scales. Composability with fine-tuned models is validated on SciERC, DocRED, and WikiEvents with Qwen3-8B and Llama-3.1-8B (RE: DR $\leq$ 0.20%, R-F1 within $\pm$0.6 pp; EE: RoV$\rightarrow$0%, AC$\leq$0.8%), but evaluation is limited to one fine-tuning method (LoRA); residual CF (Qwen 1.18%, Llama 0.46% on SciERC) traces to schema-level type ambiguity (GENERIC$\leftrightarrow$X) rather than enforcement failure. WikiEvents fine-tuning composability is preliminary (20-document test set, the full official split). All main experiments use greedy decoding; our evaluation of beam search on the full SciERC test set (551 documents) and a DocRED subset across three models confirms compatibility (Appendices E, F), but comprehensive evaluation across all decoding configurations remains future work.

Engineering considerations. Scalability benchmarks up to 200 entity types and 1,000 relations show constant 2–3% inference overhead and sub-second compilation (<0.5 s; Appendix K); however, the first-token uniqueness requirement for continuation masking may become a tokenizer-dependent bottleneck beyond this scale. Evaluation covers LoRA and full fine-tuning on 8B-class models; full fine-tuning with an untuned learning rate underperforms LoRA (R-F1 37.89 vs. 43.77 on SciERC) but confirms that constraints remain effective across fine-tuning methods (DR 2.74% $\rightarrow$ 0%; Appendix L).

Ethics Statement

This work uses publicly available benchmark datasets (SciERC, DocRED, and WikiEvents) with no personally identifiable information. No human subjects were involved in the experiments. The constrained decoding method operates on model outputs and does not introduce new privacy or safety risks beyond those inherent in the base language models. Code and data will be released upon acceptance.

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A Prompt Templates

All experiments use the same prompts across constrained and unconstrained decoding; constraints are applied at the logit level, not through prompt modification. Table 6 shows the system and user prompts for each dataset.

B Extended Related Work

Grammar-Constrained Decoding. Recent work has established efficient algorithms for restricting LLM outputs to conform to context-free grammars, making grammar-constrained decoding the de facto standard for obtaining structured output from language models. XGrammar (Dong et al., 2025) partitions the vocabulary into context-independent and context-dependent tokens and maintains persistent pushdown automata across decoding steps, achieving up to 100× speedup over prior CFG-constrained decoding methods. Outlines (Willard and Louf, 2023) reformulates constrained generation as finite-state machine transitions with offline vocabulary indexing, which enables efficient enforcement of regular-expression and JSON-schema constraints without runtime overhead per token. LMQL (Beurer-Kellner et al., 2023) introduces a declarative query language that compiles type, length, and regex constraints into token-level masks, providing a programmable interface for specifying multi-variable output structures. These systems ensure that every generated token belongs to a grammar-valid continuation, but they do not address the *distribution distortion* introduced by naive token masking. A parallel line of work tackles this problem directly: DOMINO (Beurer-Kellner et al., 2024) minimizes distributional divergence during CFG enforcement by formulating mask selection as an optimization problem, and ASAp (Park et al., 2024) provides a provably unbiased sampling algorithm that preserves the original LM distribution under grammar constraints. Together, these advances have made grammar-constrained decoding both fast and statistically principled, and provide a

SciERC — System Prompt (Unconstrained & Constrained)

```
You are an expert information extraction system for scientific texts. Extract
entities and relations from the given text.

Entity types: Task, Method, Material, Metric, OtherScientificTerm, Generic
Relation types: Used-for, Feature-of, Hyponym-of, Part-of, Compare,
Conjunction, Evaluate-for

Output a JSON object with:
- "entities": list of {"text": "...", "type": "..."}
- "relations": list of {"head": "...", "tail": "...", "type": "...",
"head_type": "...", "tail_type": "..."}

Output ONLY valid JSON, no explanation.
```

SciERC — User Prompt

```
Extract all entities and relations from this scientific text:
"{text}"
Output JSON:
```

DocRED — System Prompt (Unconstrained & Constrained)

```
You are an information extraction system. Extract all entities and relations
from the following document.

Entity types: PER, ORG, LOC, TIME, NUM, MISC

Relation types: country, instance of, performer, ... (96 Wikidata relation
types; full list omitted for space)

Output a JSON object with exactly this structure:
{"entities": [{"text": "entity name", "type": "ENTITY_TYPE"}],
 "relations": [{"head": "head entity", "tail": "tail entity", "type":
"relation name",
  "head_type": "ENTITY_TYPE", "tail_type": "ENTITY_TYPE"}]}

Output ONLY the JSON, no other text.
```

DocRED — User Prompt

```
(Document text provided inline in the user message.)
```

Table 6: Extraction prompts used in all experiments. The same prompts are used for both unconstrained and constrained decoding; OntoCoDec's constraints operate at the logit level during generation. Schema-informed prompts (§5) additionally include valid (head type, tail type) pairs per relation.

robust foundation for structured generation. However, all of these methods operate exclusively at the syntactic level: they ensure format validity (e.g., well-formed JSON, valid enum values) but cannot enforce cross-field semantic constraints such as ontology domain/range rules that depend on previously generated content. This syntactic-only limitation motivates OntoCoDec's semantic extension: our grammar layer builds on XGrammar for structural (JSON schema and enum) constraints, while a second semantic layer enforces relational type constraints that are inherently beyond the expressive power of context-free grammars.

Semantic Constrained Decoding for Code. In programming-language generation, constrained decoding has been extended beyond syntax to enforce semantic properties such as type correctness; this work demonstrates that richer-than-CFG constraints can be compiled into token-level filtering. In the text-to-SQL domain, PICARD (Scholak et al., 2021) maintains an incremental SQL parsing state that rejects tokens violating syntax or referencing non-existent database columns at each decoding step, achieving state-of-the-art results on Spider and CoSQL without modifying model weights. While mechanistically similar to OntoCoDec's token-level filtering, PICARD operates over a *fixed, fully enumerated* schema (database table and column names known before decoding begins). In contrast, OntoCoDec's entity registry is populated *during* generation as entity spans are extracted from free text; constraint sets evolve with the decoding prefix, a fundamental distinction between querying a known schema and extracting from unstructured text. Synchromesh (Poesia et al., 2022) pioneers Constrained Semantic Decoding (CSD), which enforces syntax, type, and scope constraints simultaneously during code generation from pre-trained LMs without fine-tuning; this establishes that se-

mantic constraints can be expressed as dynamic token masks conditioned on the generation prefix. MGD (Agrawal et al., 2023) uses static-analysis monitors to perform incremental type checking at decode time; SantaCoder-1.1B with MGD outperforms text-davinci-003 on repository-level code completion by ensuring that every generated identifier is type-correct with respect to the surrounding code context. Most recently, Mündler et al. (2025) compile full type systems, including generic types and subtyping, into prefix automata for token-level logits masking and reduce compilation errors by over 50%. This last system is the closest mechanistic analog to OntoCoDec, applying the same "type system $\rightarrow$ automaton $\rightarrow$ logits mask" pipeline, though in a fundamentally different domain. The key technical difference lies in the constraint structure: programming-language type systems feature deep subtyping hierarchies and parametric polymorphism, whereas IE ontology axioms specify flat domain/range sets; a relation R accepts head entities from set $\mathcal{D}(R)$ and tail entities from set $\mathcal{R}(R)$, with no subtype chains. Conversely, IE introduces a challenge absent in code generation: the constraint target (entity type) must be inferred from free-text spans rather than read from declared type annotations, so a stateful entity registry must track extracted types during autoregressive generation. These methods collectively demonstrate that semantic (type) constraints at decode time are both feasible and highly effective, but exclusively in the code domain where type systems are formally specified. OntoCoDec is the first to *automatically compile* IE ontology axioms, including domain/range constraints, into token-level decoding masks for information extraction, migrating the type-constrained decoding paradigm from code generation to IE.

Vocabulary and Type-Constrained Generation. The idea of enforcing entity-type compatibility constraints for information extraction dates back to pre-neural methods. Roth and Yih (2004) formulate joint entity and relation identification as an integer linear program (ILP) that enforces type constraints (e.g., EMP-ORG only between PER and ORG entities); this establishes the foundational principle that relational type compatibility should be a hard constraint, not a soft preference learned from data. In the autoregressive era, GENRE (De Cao et al., 2021) pioneers trie-based constrained beam search for autoregressive entity retrieval, restricting each decoding step to tokens consistent with valid entity names from a predefined catalog. GenIE (Josifoski et al., 2022) extends this approach to end-to-end information extraction with state-based constraints that enforce valid triple structures, achieving strong performance on closed-world knowledge base population, though it requires task-specific fine-tuning on the target schema. AdaKGC (Ye et al., 2023) introduces schema-constrained dynamic decoding with trie-based prefix trees that adapt to the target schema, enforcing that generated entity and relation names belong to a predefined inventory, a vocabulary-level constraint that ensures name validity but does not verify type compatibility between a relation and its arguments. GCD (Geng et al., 2023) provides a unified grammar-constrained decoding framework for structured NLP tasks including IE, introducing input-dependent grammars that adapt constraints based on the source text. GCD is the closest precursor to OntoCoDec in applying logits masking to IE, though limited to syntactic constraints. NeuroLogic Decoding (Lu et al., 2021) expresses lexical inclusion and exclusion constraints as predicate logic formulas enforced during beam search, offering a general mechanism for token-level control that can encode vocabulary membership but not relational type rules. These approaches constrain *which tokens* can appear (vocabulary membership) but not *which type combinations* are valid for a given relation. OntoCoDec's semantic layer enforces relational constraints (e.g., that the head entity of relation R must belong to type set $\mathcal{D}(R)$) which cannot be expressed as static vocabulary restrictions because the valid set depends on the relation type generated earlier in the same sequence. While Roth & Yih's ILP formulation shares the same type-compatibility motivation, it operates as a post-hoc reranker over discrete candidates; OntoCoDec integrates enforcement directly into autoregressive decoding, preventing violations rather than filtering them.

Structural Constraints via Training and Prompting. An orthogonal approach encodes IE output structure through model training or prompt design rather than decoding-time enforcement, relying on learned representations to produce schema-compliant outputs. UIE (Lu et al., 2022) defines a Structured Extraction Language and trains a unified text-to-structure model across 13 IE datasets, relying on large-scale pre-training to internalize output format constraints. GoLLIE (Sainz

et al., 2024) expresses annotation guidelines as Python dataclass definitions and fine-tunes Code-LLaMA to generate typed code, achieving strong zero-shot transfer to unseen IE schemas through the implicit type structure of programming languages. KnowCoder (Li et al., 2024) takes this further by encoding 29K entity types from Wikidata as a Python class hierarchy, training a Code-LLM to generate structured extraction code that implicitly respects type inheritance, the most comprehensive schema-as-code approach to date. CodeIE (Li et al., 2023) similarly reformulates IE as code generation, using the structural regularity of programming languages to reduce output format errors. ADELIE (Qi et al., 2024) uses SFT and DPO to align LLMs on IE tasks, achieving state-of-the-art results among open-source models while preserving general capabilities, though compliance with ontological constraints remains a learned soft preference rather than a guarantee. In the supervised setting, joint entity-relation architectures such as SpERT (Eberts and Ulges, 2020) and PFN (Yan et al., 2021) enforce type consistency implicitly through shared representations: SpERT classifies entity spans and relation pairs in a single BERT pass with span-level negative sampling, while PFN decomposes feature encoding into entity-specific, relation-specific, and shared partitions via gating mechanisms. These models achieve strong performance but are limited to fixed, predefined schemas and cannot generalize to unseen ontologies without retraining, precisely the ontology-portable, training-free scenario that OntoCoDec targets. EnriCo (Zaratiana et al., 2024) applies Answer Set Programming (ASP) constraints during decoding of joint entity-relation extraction models, enforcing structural type compatibility at generation time. This is a decoding-time approach, though with model-specific, manually authored constraints rather than ontology-driven compilation. Training-based methods provide *soft* structural guidance: the model learns to prefer valid outputs but cannot guarantee compliance at inference time, as demonstrated by non-trivial violation rates even in fine-tuned systems. Importantly, training-based and decoding-time approaches are complementary rather than competing: a system like GoLLIE could be combined with OntoCoDec's constraint layers, using fine-tuning to maximize the probability of valid outputs while decoding-time enforcement eliminates residual violations. OntoCoDec provides *hard* decoding-time enforcement that is training-free, automatically compiled from ontology schemas, and composable with any of these training-based approaches.

Post-hoc Ontology Validation. Rather than constraining generation, several methods validate LLM outputs against ontological rules after generation, applying a detect-then-fix paradigm. ODKE+ (Khorshidi et al., 2025) dynamically generates ontology snippets containing domain/range constraints as prompt context, achieving 98.8% precision across 195 predicates at web scale, but constraint adherence relies on prompt-level guidance, not enforcement, leaving residual violations unaddressed. SPIRES/OntoGPT (Caufield et al., 2024) provides an ontology-based extraction framework that grounds LLM outputs against OBO ontologies post-generation: it validates extracted entities and relations through ontology lookup and corrects mismatches in a second pass, the prompt-then-validate paradigm in the biomedical domain. OntoLogX (Cotti et al., 2025) applies SHACL validation to LLM-generated knowledge graphs, iteratively re-prompting the LLM to correct constraint violations and reducing them by approximately 10× across multiple correction rounds. This demonstrates the detect-then-fix paradigm's effectiveness but also its inherent multi-pass cost. Upadhyay et al. (2023) inject entity-type knowledge as constraints into Open IE via both constrained inference and constraint-aware training, improving F1 by 29% on CaRB, though their constrained inference operates over a discrete candidate set rather than during autoregressive generation. Regino and dos Reis (2025) propose a multi-stage LLM pipeline for validating RDF triple insertions against RDFS domain/range constraints, where each validation step requires a separate LLM call, effective but computationally expensive at scale. RELATE (Olasunkanmi et al., 2025) addresses the alignment gap between LLM-extracted biomedical relations and standardized ontology predicates through a three-stage pipeline: ontology preprocessing with predicate embeddings, similarity-based retrieval via SapBERT, and LLM-based reranking with negation handling, achieving 94% accuracy@10 on ChemProt. While RELATE demonstrates effective ontology grounding for relation extraction, its constraints are applied *post-extraction* through retrieval and reranking rather

than during generation, and it requires domain-specific embedding models trained on biomedical corpora. OntoCoDec enforces ontology constraints *during* autoregressive decoding and requires no additional trained components beyond the base LLM. Concurrently, Zhou et al. (2026) compile OWL ontologies into executable tool interfaces within The World Avatar framework; LLM agents must invoke ontology-aware tools that enforce semantic constraints during knowledge graph construction from scientific text. This "ontology-to-tools" approach shares OntoCoDec's philosophy of compiling formal ontological knowledge into runtime enforcement mechanisms, but operates at the tool-call level (constraining which API functions the agent invokes) rather than at the token level (constraining which tokens the decoder produces), and requires an agentic multi-step workflow rather than single-pass autoregressive generation. Mehenni and Zouaq (2024) introduce ontology-constrained beam scoring for clinical summarization, applying soft biases (not hard masks) during beam search to steer outputs toward ontology-compliant terms, a middle ground between unconstrained generation and hard enforcement that reduces but cannot eliminate violations. Post-hoc validation detects violations after generation and requires rejection or correction cycles that waste compute. In our setting, 15.37% of generated triples contain domain/range violations under unconstrained decoding, a substantial waste of generation effort. OntoCoDec *prevents* violations during decoding, redirecting probability mass to valid alternatives rather than discarding invalid outputs, with only marginal overhead from logits masking compared to the cost of regenerating or correcting violated triples. Soft-constraint approaches such as ontology-weighted beam scoring reduce violation frequency but cannot eliminate it, whereas OntoCoDec substantially reduces violations (DR↓94% on SciERC, ↓45–90% on DocRED depending on model).

Quality Impact of Constrained Decoding. A growing body of work raises concerns that constrained decoding degrades LLM generation quality, particularly for reasoning-intensive tasks where format enforcement interferes with chain-of-thought processes. Tam et al. (2024) conduct a comprehensive evaluation showing that rigid format constraints (JSON, XML) harm performance on math and multi-hop QA benchmarks but *benefit* classification-style tasks such as slot filling, a distinction directly relevant to IE, which is fundamentally an extraction (classification-like) task rather than a reasoning task. CRANE (Banerjee et al., 2025) provides theoretical analysis of why restricting outputs to narrow grammars suppresses chain-of-thought reasoning by eliminating intermediate reasoning tokens, and proposes an alternating strategy between unconstrained reasoning phases and constrained output phases to mitigate this effect. Schall and de Melo (2025) conduct a systematic study across 11 models and multiple benchmarks, finding 10–30% semantic correctness degradation on reasoning tasks under structural constraints, with larger models showing greater absolute degradation. Zhou (2026) identify "structure snowballing," a failure mode where format compliance cognitive load traps models in repetitive structural patterns rather than improving reasoning during self-correction iterations. These findings establish a legitimate concern for constrained decoding in reasoning-heavy applications. However, information extraction occupies a fundamentally different regime: the model extracts entities and relations from input text rather than generating novel reasoning chains, and Tam et al. (2024)'s own results show that extraction-style tasks are robust to, or even benefit from, format constraints. OntoCoDec addresses this concern empirically: domain/range violations decrease by 94% on SciERC (15.37%→0.93%) and 45–90% on DocRED across two models, with F1-neutral impact on SciERC (+0.13pp, p=0.58) and significant R-F1 gains on DocRED (Qwen $p<0.001$, Llama p=0.013). Critically, OntoCoDec's constraints target only type tokens (entity types, relation labels), leaving free-text spans (entity surface forms and their boundaries) entirely unconstrained, which minimizes the distribution distortion that causes quality degradation in reasoning tasks. A two-pass variant (S2), explored in ablation (Section 5), separates extraction and constraint enforcement into distinct passes, analogous to CRANE's alternating strategy; however, our primary single-pass approach already avoids reasoning interference by constraining only type tokens while leaving free-text spans unconstrained.

Our Approach. OntoCoDec occupies a unique position in this landscape: it is the first system to *automatically compile* IE ontology schemas, including domain/range axioms, into token-level semantic masks, extending grammar-constrained decoding from syntactic to semantic enforce-

ment without manually authored rules. Unlike vocabulary-constrained methods that restrict individual tokens to a fixed set, OntoCoDec enforces *cross-field relational constraints* through a stateful entity registry that tracks extracted types during autoregressive generation; constraint evaluation is dynamic and depends on previously decoded content. This approach is orthogonal to supervised IE methods and applicable to any autoregressive LM without model modification or additional training. Unlike training-based IE systems, OntoCoDec is training-free and substantially reduces type violations at inference time (DR↓94% on SciERC, ↓45–90% on DocRED); unlike post-hoc validation, it prevents violations rather than detecting them, redirecting probability mass to valid alternatives with significant R-F1 gains on DocRED ($p \leq 0.013$) and F1-neutral impact on SciERC. The two constraint layers, grammar (structural) and semantic (relational), compose via bitwise AND with an enabling dependency: the grammar layer ensures every generated type token belongs to the valid vocabulary, which enables the semantic layer to look up and enforce domain/range rules for that type.

C Extended Discussion

CF Dual Mechanism. CF violations arise from two distinct mechanisms: (1) *Registry override*: the semantic constraint forces a domain-compatible type that conflicts with the entity's previously registered type. The entity appears in multiple relations with different type requirements, and the per-relation mask selects the locally valid type without consulting prior assignments. (2) *Grammar-first fallback* (§3.3): when the semantic mask conflicts with all grammar-valid tokens, the semantic layer is dropped entirely, allowing the grammar layer to generate any structurally valid type, which may differ from the entity's registered type.

These mechanisms have sharply different model profiles. Qwen3-8B triggers zero grammar-first fallbacks across all experiments, attributing 100% of its CF violations to registry override. Llama-3.1-8B, by contrast, triggers 5,882 fallbacks on SciERC (10.7 per document) and 57,825 on DocRED (63.8 per document), making grammar-first fallback the dominant CF source. This asymmetry is consistent with the cross-model CF gap: Qwen achieves CF = 2.57% on DocRED vs. Llama's 12.02%, despite both using identical constraint configurations. The difference traces to Llama's less calibrated type predictions, which more frequently conflict with the semantic mask's valid set, triggering fallback.

Fallback Behavior and Effective Coverage. The two-layer architecture produces two distinct fallback mechanisms with different semantic consequences. *Grammar-first fallback* (36.6% of type-bearing positions on SciERC) occurs when $S(t) \cap G(t) = \emptyset$: the semantic layer deactivates as a safety valve, preserving structural validity at the cost of potential type violations. These events account for the residual DR that persists after constraint enforcement. *Registry-consistent fallback* (4.5%) occurs when the entity's registered type conflicts with domain/range constraints; the system defaults to the registered type for cross-triple consistency. Together, these mechanisms produce the characteristic profile of composable constraint systems: residual DR traces to grammar-first fallback events, while CF violations arise from registry override events where per-relation enforcement assigns conflicting types.

We define *effective semantic constraint coverage* as $1 - f_{\text{gf}}$, where f_{gf} is the grammar-first fallback rate. On SciERC, Llama-3.1-8B shows $f_{\text{gf}} = 36.6\%$ (6,237 of 17,024 mask calls), yielding 63.4% effective coverage. On DocRED, by contrast, Qwen3-8B shows $f_{\text{gf}} = 0.08\%$ (46 of 54,859), yielding 99.9% coverage. The large discrepancy is primarily a tokenization artifact: SciERC entity types are multi-token (e.g., OTHERSCIENTIFICTERM, averaging 2.87 mask calls per type field), and continuation tokens necessarily conflict with the semantic mask, inflating the per-mask-call fallback rate. DocRED types (PER, ORG, LOC, TIME, NUM, MISC) are all single-token, eliminating this inflation. This metric directly quantifies the enabling dependency (§3.3): it measures the fraction of decoding steps where both constraint layers are jointly active, providing a fine-grained view beyond aggregate violation rates. Qwen3-8B, with zero grammar-first fallbacks, achieves 100% effective coverage on SciERC, consistent with its lower residual DR (0.93% vs. Llama's 0.08%).

CF-DR Trade-off Analysis. CF violations are the most prominent residual limitation of OntoCoDec under zero-shot decoding (9.90% on SciERC, up to 12.02% on Llama DocRED). We treat this honestly: per-relation independent enforcement cannot eliminate CF without trading it against DR. Table 7 reports a pilot ablation in which we replace the grammar-first fallback with a CF-priority

Configuration	DR (%)	CF (%)	DR+CF	f_{gf} (%)
Full S1 (baseline)	0.93	9.90	10.83	—
CF fallback	14.78	0.04	14.82	36.64
Δ	+13.85	−9.86	+3.99	—

Table 7: CF-DR trade-off pilot on SciERC (Qwen3-8B, 551 documents; `w1_cf_fallback_scierc_pilot`). CF fallback eliminates CF but re-routes violations to DR, increasing total violations (DR+CF) by 3.99 pp. f_{gf}: grammar-first fallback rate at type-bearing positions.

fallback that forces the entity's registered type when it conflicts with $\sigma(r)$.

The CF fallback variant nearly eliminates CF (9.90% → 0.04%, −9.86 pp) but at the cost of a 15.9× increase in DR (0.93% → 14.78%, +13.85 pp). This is not a net improvement: total violations (DR+CF) rise from 10.83% to 14.82% (+3.99 pp), so the change *re-routes* violations rather than reducing them. The mechanism is mechanical: each token position with conflicting masks must commit to one violation type; when the CF fallback forces the system to emit the entity's registered type even though it lies outside $\sigma(r)$, the resulting triple satisfies cross-entity consistency but violates domain/range. The grammar-first fallback is triggered at 36.64% of type-bearing positions; this conflict is structural, not a tunable hyperparameter.

We interpret this as a fundamental tension in composable constraint systems with *per-relation independent* enforcement: enforcement priority can be allocated between DR and CF but cannot independently minimize both. Within this design, the current full S1 configuration sits near the Pareto frontier of the trade-off, achieving 0.93% DR at the cost of 9.90% CF. The direction of the trade-off is consistent beyond SciERC/Qwen3-8B: across all six zero-shot configurations (3 models × 2 datasets), DR and CF are inversely associated (Spearman $\rho = -0.77$, n=6), with no setting showing both high DR and high CF simultaneously (Table 4). The pattern is a structural consequence of per-relation enforcement with grammar-first fallback, not a model- or dataset-specific artifact; we select SciERC/Qwen3-8B for detailed ablation because it exhibits the highest CF (9.90%), making the trade-off most visible. While the inverse association holds across all configurations, the absolute trade-off magnitude may differ in settings with higher baseline CF, such as Llama DocRED (CF = 12.02%), where the larger CF headroom could yield proportionally larger shifts under alternative fallback priorities. The CF residual is concentrated in schema ambiguity rather than enforcement failure: 63.3% of CF violations on SciERC involve the catch-all GENERIC type (Table 8), which competes with every content type depending on anaphoric vs. referential use, a distinction the per-slot decoder cannot resolve without document-level coreference.

Two complementary paths can lower CF without re-routing it to DR. (i) *Reduce the baseline conflict rate.* Composing OntoCoDec with LoRA fine-tuning on SciERC reduces CF by 88% relative to the zero-shot setting (9.90% → 1.18%; Table 2), because the fine-tuned model's per-slot type predictions agree more often with the entity registry, so the per-relation mask and the registry intersect more frequently. The remaining 1.18% still traces to GENERIC↔X schema ambiguity, so this path is empirically validated on one dataset and one fine-tuning method but does not generalize without further evidence. (ii) *Replace per-relation enforcement with global entity-level type consensus during decoding.* This is outside the scope of OntoCoDec's forward-only autoregressive design and is left to future work.

Dual-Factor Quantitative Analysis. Figure 1 plots residual DR against the product $\bar{\kappa} \times \text{RV}_{\text{baseline}}$ for model×dataset configurations. For *calibrated* models (Qwen3-8B, Llama-3.1-8B; unconstrained ET<3% on both datasets), a linear fit yields $y = 0.366x - 0.138$ with $R^2 = 0.886$ (n=4, $F(1,2) = 15.54$, $p \approx 0.058$); with only four data points (df = 2) and $p > 0.05$, this constitutes a descriptive characterization rather than a confirmatory statistical test. The strong fit supports the dual-factor pattern from §4: residual DR is governed by the product of schema tightness and baseline violation rate. Mistral-7B (unconstrained ET=7.58% on DocRED) deviates from this pattern: its residual DR on DocRED (33.21%) substantially exceeds the linear prediction. When model calibration is insufficient (ET>5% on either dataset), residual DR is dominated by entity misclassification rather than constraint-search failure. This motivates a principled exclusion criterion: models with unconstrained ET>5% on either dataset fall outside the regime where the tightness×RV product is the primary DR driver. The two DocRED points (high $\bar{\kappa}$=0.842) separate cleanly from the two SciERC

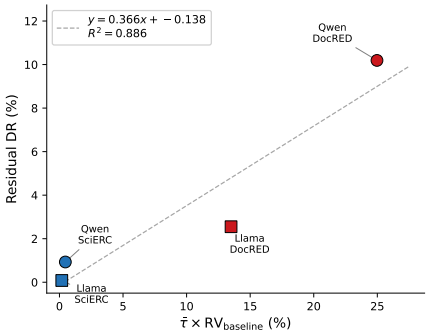


Figure 1: Residual DR vs. $\bar{\kappa} \times \mathrm{RV}_{\mathrm{baseline}}$ across four model×dataset configurations. Linear fit (calibrated models): $y = 0.366x - 0.138$, $R^2 = 0.886$; Mistral deviates due to entity misclassification.

points (low $\bar{\kappa}$=0.333), and within each dataset the model with higher baseline RV shows higher residual DR.

Dataset-Dependent F1 Impact. The divergence in F1 significance between DocRED (+0.50 pp, $p < 0.001$) and SciERC (+0.13 pp, $p = 0.58$) traces to a single mechanism: grammar-layer elimination of out-of-vocabulary relation names. DocRED's baseline RV of 29.66% produces 5,189 OOV predictions counted as false positives; eliminating them yields a significant precision gain. SciERC's baseline RV is only 1.39% (24 OOV predictions), too few to produce a detectable F1 effect. Llama-3.1-8B's lower baseline RV (16.03%) yields a smaller but directionally consistent improvement (+0.29 pp vs. +0.50 pp); F1 gain magnitude is proportional to baseline OOV rate.

Practical Significance of Zero-Shot F1. Our reported relation F1 of approximately 8% on both SciERC and DocRED reflects the inherent difficulty of zero-shot closed IE, where the model receives no task-specific training examples. For reference, supervised state-of-the-art systems on SciERC achieve approximately 49–53% relation F1 with full training data (Zhong and Chen, 2021; Ye et al., 2022). Constraint enforcement operates largely independently of the extraction model's intrinsic capability: it restricts the output space to ontologically valid triples regardless of the underlying model's accuracy. Even after LoRA fine-tuning, residual DR of 1–2% persists on tight-constraint relations (Table 2); constraints remain necessary at deployment-grade quality. In our evaluation, F1 serves as a *control variable* verifying that constraint enforcement does not degrade extraction quality, not as an optimization target. The contribution of this work is the constraint enforcement method itself; the zero-shot setting stress-tests constraint robustness under maximum model uncertainty, while composability with fine-tuning (§5.1) validates practical deployment value.

Decoding-Time vs. Post-Hoc Enforcement. Post-hoc filtering (generating triples without constraints, then removing schema-violating predictions) yields higher precision and consequently higher F1 than OntoCoDec on both datasets (SciERC 8.63 vs. 8.36; DocRED 9.61 vs. 8.23). However, this precision gain comes at substantial cost. On DocRED, post-hoc filtering discards 48.1% of all predicted triples, including 127 true positives (11.2% of correct predictions); recall drops from 9.46% to 8.40%. Entity type violations remain unresolved (ET = 2.47% vs. OntoCoDec's 0%), as filtering targets only domain/range and vocabulary violations. OntoCoDec's decoding-time approach avoids this information loss by guiding generation toward compliant outputs rather than discarding non-compliant ones, preserving both recall and entity type validity. The two strategies are complementary: post-hoc filtering can serve as an additional safety net for residual violations that decoding-time enforcement does not fully resolve.

Cross-Field Violation Distribution. Table 8 reports the complete distribution of conflicting-type pairs for SciERC full S1 (245 CF violations, all pairwise; no mention triggered ≥3 types). Conflicts involving the GENERIC type account for 63.3% (155/245) of all violations. GENERIC is SciERC's anaphoric/placeholder label ("this approach", "the method"), which competes with every content type depending on whether the mention is used anaphorically or referentially, a distinction the per-slot constrained decoder cannot resolve without document-level coreference.

The pattern generalizes across models and datasets: on Llama-3.1-8B SciERC, GENERIC↔OTHERSCIENTIFICTERM remains the top pair (53/162, 32.7%); on DocRED, the analogous catch-all type MISC drives 76.6% of Qwen's CF violations. Under unconstrained decoding, CF rates collapse to ≤1% on both datasets; CF violations are a side-effect of per-slot type commitment rather than a model deficiency.

Per-Relation DR Elimination under Fine-Tuning. Table 9 presents the per-relation break-

Conflicting Type Pair	Count	%
Generic ↔ OtherScientificTerm	85	34.7
Generic ↔ Material	35	14.3
Generic ↔ Metric	30	12.2
Method ↔ Task	20	8.2
Material ↔ Method	17	6.9
OtherScientificTerm ↔ Task	12	4.9
10 further pairs (each $\leq$3.3%)	46	18.8
Total	**245**	100.0

Table 8: CF violation distribution on SciERC full S1 (Qwen3-8B, 551 documents). All 245 violations are pairwise conflicts. Conflicts involving GENERIC account for 63.3%.

		LoRA Unconstr.		LoRA + S1	
Relation	κ	DR (%)	n	DR (%)	n
Evaluate-for	0.361	13.13	99	**0**	98
Part-of	0.417	5.77	52	**0**	52
Compare	0.528	5.00	40	**0**	40
Conjunction	0.306	1.31	153	**0**	153
Feature-of	0.278	1.35	74	**0**	74
Used-for	0.028	0.16	634	**0**	635
Hyponym-of	0.417	0	—	**0**	—

Table 9: Per-relation DR elimination on SciERC with LoRA fine-tuning (Qwen3-8B). κ: constraint tightness; n: predicted triple count. Used-for gains one triple from Evaluate-for redirection.

down of DR elimination when OntoCoDec is composed with LoRA fine-tuning on SciERC. Fine-tuning alone eliminates DR on four of seven relations but leaves non-zero violations on the three tightest-constraint relations: EVALUATE-FOR (κ=0.361, 13.13%), PART-OF (κ=0.417, 5.77%), and COMPARE (κ=0.528, 5.00%). Adding OntoCoDec constraints eliminates all remaining violations across all relations. One EVALUATE-FOR triple (99 → 98) is redirected to USED-FOR (634 → 635) by token-level masking: the head entity's registered type falls outside EVALUATE-FOR's domain, and the semantic mask steers generation to the nearest valid relation.

Fine-Tuned Four-Way Ablation. Table 10 presents the four-way constraint ablation on fine-tuned models across both datasets, isolating each layer's contribution when the base model has already internalized schema knowledge.

The grammar layer is effectively a no-op for fine-tuned models across all four settings: RV and ET are already 0% without constraints, and grammar-only DR change is <0.2 pp (4/4 settings). The semantic layer is the sole effective DR defense under fine-tuning, independently reducing DR by 39–72% across all settings (SciERC: Qwen −69%, Llama −41%; DocRED: Qwen −72%, Llama −39%). Adding both layers achieves near-complete elimination (DR $\leq$ 0.20% in all 4 settings): even after fine-tuning, the semantic layer's pair-based enforcement captures violations that training alone cannot eliminate, specifically on tight-constraint relations where the valid type-pair space is small. F1 cost is negligible: the maximum R-F1 decrease across all 16 model×dataset×mode combinations is −0.30 pp (Qwen DocRED, full vs. unconstrained), attributable to parse failure increase (7.11% → 8.52%) rather than constraint-induced extraction degradation.

Dataset	Model	Mode	DR↓	RV↓	ET↓	CF	PF	E-F1↑	R-F1↑
SciERC	Qwen3-8B	Unconstrained	1.95	0.00	0.00	0.00	0.00	67.05	43.86
		Grammar	1.90	0.00	0.00	0.00	0.00	67.05	43.86
		Semantic	0.60	0.00	0.00	0.18	0.00	67.05	43.69
		Full S1	**0**	0.00	0.00	1.18	0.00	67.05	43.77
	Llama-3.1-8B	Unconstrained	1.11	0.00	0.00	0.00	0.00	67.76	45.45
		Grammar	1.11	0.00	0.00	0.00	0.00	67.76	45.45
		Semantic	0.65	0.00	0.00	0.10	0.00	67.76	45.47
		Full S1	**0.09**	0.00	0.00	0.46	0.00	67.76	45.47
DocRED	Qwen3-8B	Unconstrained	0.76	0.27	0.00	0.12	7.11	82.15	40.86
		Grammar	0.61	0.00	0.00	0.11	7.01	82.19	40.85
		Semantic	0.21	0.23	0.00	0.27	7.52	82.15	40.89
		Full S1	**0.01**	0.00	0.00	0.41	8.52	81.43	40.56
	Llama-3.1-8B	Unconstrained	0.49	0.15	0.00	—	2.30	83.93	43.72
		Grammar	0.51	0.00	0.00	0.12	2.30	84.04	43.77
		Semantic	0.30	0.15	0.00	0.16	2.40	84.02	43.90
		Full S1	**0.20**	0.00	0.00	0.26	2.51	85.23	44.23

Table 10: Four-way constraint ablation on fine-tuned models (LoRA, 2 models × 2 datasets). DR/RV/ET/CF: violation rates (%); PF: parse failure rate (%); F1 ×100. Grammar layer is a no-op post-fine-tuning (DR change <0.2 pp in all 4 settings; RV and ET already 0% without constraints). Semantic layer is the sole effective DR defense, providing 39–72% reduction. F1 cost is negligible: max ΔR-F1 = −0.30 pp (Qwen DocRED full vs. unconstrained).

Residual DR Error Analysis. Across all fine-tuned settings (2 models × 2 datasets × full S1), only 36 domain/range violations remain. Manual inspection reveals four root causes:

1. *Entity type errors* (39%): the LLM assigns an incorrect entity type (e.g., labeling a geographic entity as LOC when the gold standard requires MISC), causing the semantic mask to permit a domain/range-incompatible relation.
2. *Schema coverage gaps* (33%): the frequency-based constraint table, constructed from training-set co-occurrence statistics, does not cover long-tail type pairs that are ontologically valid but rare. These violations are irreducible by decoding-time constraints alone.
3. *Incorrect relation selection* (17%): the model

Setting	Model	Dataset	Reject%	R-F1$_{ph}$	R-F1$_{cd}$	Winner
ZS	Qwen	SciERC	16.8	8.63	8.36	ph
	Llama	SciERC	13.9	2.71	2.54	ph
	Mistral	SciERC	20.3	3.94	3.67	ph
	Qwen	DocRED	48.1	9.61	8.23	ph
	Llama	DocRED	42.2	7.44	6.27	ph
	Mistral	DocRED	**73.7**	3.85	**5.52**	cd
FT	Qwen	SciERC	1.95	43.94	43.77	ph
	Llama	SciERC	1.11	45.23	45.47	cd
	Qwen	DocRED	1.03	41.11	40.56	ph
	Llama	DocRED	0.64	43.92	44.23	cd

Table 11: Post-hoc filtering (ph) vs. constrained decoding (cd). In zero-shot settings, post-hoc F1 is higher in 5/6 groups (precision-dominated), but constrained decoding preserves higher recall in all 6/6 groups. At reject rate >70% (Mistral DocRED), post-hoc F1 collapses below constrained. Under fine-tuning, rejection rates drop to 0.6–2.0% and the two strategies are equivalent ($\Delta \leq 0.55$ pp; each wins two groups). F1 $\times 100$.

selects a semantically inappropriate relation whose domain/range constraints are incompatible with the correctly typed entities.

4. *Direction reversal* (11%): the model assigns the correct relation but inverts head/tail roles, placing entities in positions that violate the relation's asymmetric domain/range specification.

Two dataset-specific patterns emerge. Outside Llama DocRED, all residual violations (3 of 36, spanning Llama SciERC and Qwen DocRED) trace exclusively to schema coverage gaps: the model's entity typing is accurate, and the constraint system has reached its theoretical ceiling given the frequency-based constraint table. On DocRED, Llama-3.1-8B contributes 33 of 36 residual violations (92%), with entity type errors as the dominant cause (14/33, 42%). This asymmetry reflects Llama's weaker fine-tuned entity typing on DocRED's larger ontology (96 relations, 7 entity types) compared to Qwen, which achieves near-zero residual. The implication is twofold: (i) the irreducible floor from schema gaps is small ($\sim$3 violations across 4 settings), so the constraint system operates near its theoretical optimum; (ii) residual violations from entity typing errors are addressable through stronger base models or improved entity typing modules, without modifying the constraint architecture.

Post-Hoc Filtering: Extended Comparison. Table 11 presents the full comparison across all six zero-shot and four fine-tuned configurations.

The crossover point occurs when post-hoc rejection exceeds $\sim$70%: Mistral DocRED rejects 73.7% of predicted triples, discarding enough true positives that constrained decoding's recall advantage dominates (+1.67 pp). OntoCoDec's advantage thus scales with violation severity: precisely when constraints are most needed, post-hoc filtering is most destructive. Under fine-tuning, both strategies converge because few violations remain to filter or constrain ($\Delta \leq 0.55$ pp across all four FT groups, with each strategy winning two).

Zero-Shot Runtime Profiling. On SciERC (Qwen3-8B, zero-shot greedy, NVIDIA RTX PRO 6000 Blackwell Server Edition), unconstrained inference averages 4.163 s/doc and Full S1 constrained decoding averages 4.710 s/doc (+13.1% overhead). On DocRED (same configuration), unconstrained inference averages 19.34 s/doc and Full S1 averages 19.81 s/doc (+2.43% overhead). The lower relative overhead on DocRED reflects amortization: the fixed per-token constraint cost is spread over longer generation sequences. Grammar compile is a one-time 0.044 s cost (amortized to zero).

Fine-Tuned Latency Profiling. Under LoRA fine-tuning (same hardware and dataset), unconstrained inference averages 2.612 s/doc and Full S1 constrained decoding averages 2.873 s/doc (+10.0% overhead, n=551). Fine-tuned models generate shorter, more structured outputs than zero-shot models (2.612 s vs. 4.163 s/doc unconstrained); absolute generation time drops by 37%. The relative overhead (+10.0%) is slightly lower than zero-shot (+13.1%), consistent with shorter sequences requiring fewer constraint checks per document.

Relation Matching Protocol. Relation F1 evaluation uses fuzzy normalization to handle model-specific formatting variants (e.g., underscores vs. spaces, abbreviations). A predicted relation r_p matches gold relation r_g if: (1) exact string match after case-insensitive normalization, or (2) token overlap >80% *and* edit distance ≤ 2. This protocol affects primarily Mistral, which produces underscore-separated variants (e.g., "head_of_government" for "head of government"). On Mistral DocRED, fuzzy matching contributes 98 additional TPs (+0.84 pp R-F1; strict R-F1 = 3.50% vs. fuzzy 4.34%). For Qwen and Llama under grammar constraints, the grammar layer enforces exact relation names, making fuzzy matching irrelevant (0 additional matches). All F1 figures in the main text use fuzzy normalization

Table 12: Violation rates (%) under unconstrained and schema-informed prompting (Qwen3-8B, zero-shot).

Dataset	Prompt	DR↓	RV↓	ET↓	CF↓
SciERC	Unconstrained	15.4	1.4	0.4	0.1
	Schema-informed	5.8	0.3	**0.0**	—
DocRED	Unconstrained	18.5	29.7	2.5	0.1
	Schema-informed	12.8	18.4	0.2	0.2

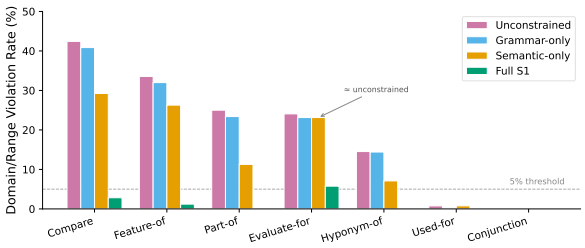


Figure 2: Per-relation DR violation rates on SciERC (Qwen3-8B, unconstrained). Rates correlate with constraint tightness $\kappa(r)$.

consistently across all models and conditions.

Per-Relation Violation Landscape. Table 12 compares violation rates under unconstrained and schema-informed prompting. Figure 2 shows the per-relation DR distribution on SciERC.

WikiEvents Dataset Details. WikiEvents (Li et al., 2021) is a document-level event argument extraction benchmark grounded in the DARPA KAIROS ontology. It defines 56 event types, 80 argument roles, and 27 entity types, with 245 (event_type, role)→allowed_entity_types constraints. The grammar layer targets EV (event vocabulary), RoV (role vocabulary), and ET; the semantic layer enforces AC (argument constraint) via role-conditioned masking, analogous to RE domain/range enforcement.

Constraint Tightness in WikiEvents. WikiEvents role-level tightness (Eq. 3, with $|\mathcal{E}|$=27) has mean $\bar{\kappa}$=0.82 and median 0.89; 84% of roles have $\kappa \geq 0.80$ (4 or fewer valid entity types out of 27). The tightness-violation pattern extends to event extraction: roles with $\kappa \geq 0.90$ exhibit the highest AC rate (50%), compared to 21% for medium-tightness ($0.80 \leq \kappa < 0.90$) and 30% for loose ($\kappa < 0.80$) roles; this is consistent with the RE observation that constraint tightness correlates with violation frequency. However, the correlation is modulated by entity type prototypicality: semantically intuitive roles (e.g., Transportation.Vehicle→VEH) show 0% AC despite κ=0.963, while roles that require rare KAIROS-specific types (e.g., Crime→CRM, Sentence→SEN) show near-100% AC because LLMs default to common types (PER, ORG). The loose group's elevated rate (30%) is driven by high-frequency roles such as Attack.Target (126 args, 34% AC, κ=0.59), where systematic model errors outweigh the permissive constraint space.

D Per-Relation Violation Analysis (Llama-3.1-8B)

Table 13 presents the per-relation domain–range violation breakdown for Llama-3.1-8B on DocRED under full S1 constraints. The Llama baseline exhibits an RV rate of 16.03%, substantially lower than Qwen's 29.66%, which indicates stronger baseline vocabulary compliance. Full S1 completely eliminates RV violations and reduces the residual DR rate to 2.55% (418/16,368 triples), with only three relations exceeding 10%.

Table 13: Top 10 per-relation DR violation rates for Llama-3.1-8B on DocRED under full S1 constraints. Relations with >20 triples and 0% DR (e.g., P131, P206, P361, P170, P86) are omitted.

Relation	PID	Triples	DR Violations	DR Rate (%)
dissolved/demolished	P576	73	24	32.9
followed by	P156	117	24	20.5
unemployment rate	P1198	15	3	20.0
end time	P582	130	21	16.2
instance of	P31	290	37	12.8
product/material produced	P1056	98	11	11.2
location of formation	P740	81	8	9.9
head of state	P35	41	3	7.3
applies to jurisdiction	P1001	637	45	7.1
conflict	P607	453	30	6.6

Compared with Qwen, which retains a 10.21% residual DR under full S1 concentrated in P1198 (83% fallback) and P171 (78% fallback), Llama's residual DR on the same relations is markedly lower (P1198 reaches only 20.0%) and no relation-level fallback concentration comparable to Qwen is observed. The substantially lower residual DR for Llama (2.55% vs. Qwen's 10.21%) on the same ontology confirms that constraint tightness alone does not determine enforcement effectiveness: the model's baseline vocabulary compliance (RV) modulates the fraction of triples eligible for semantic-layer enforcement.

Table 14: Constraint robustness under nucleus sampling (T=0.7, p=0.9) vs. greedy decoding (Qwen3-8B, full S1 constraints).

Dataset	Decoding	ET%	RV%	DR%	CF%	E-F1	R-F1
SciERC (551)	Greedy	0	0	0.93	9.90	30.98	8.36
	Sampling	0	0	0.75	9.90	31.28	7.42
DocRED (114†)	Greedy	0	0.03	10.21	2.57	59.37	8.23
	Sampling	0	0	10.31	12.68	57.05	7.92

†Subset of 114 documents; full DocRED sampling requires ∼300h due to extended generation under high-temperature sampling with complex schemas.

Table 15: Beam search (k=4) vs. greedy decoding under full S1 constraints. SciERC evaluated on full test set (551 documents); DocRED evaluated on matched 50-document subsets (same documents for greedy and beam). ET = entity type, RV = relation vocabulary, DR = domain/range, CF = cross-field consistency violation rates (%); F1 ×100.

Model / Dataset	Decoding	ET	RV	DR↓	CF↓	E
Qwen3-8B / SciERC (551)	Greedy	0	0	0.93	9.90	
	Beam (k=4)	0	0	**0**	**0.53**	
Qwen3-8B / DocRED (50)	Greedy	0	0	8.69	2.08	
	Beam (k=4)	0	0	**0**	**0.55**	
Llama-3.1-8B / SciERC (551)	Greedy	0	0	0.08	6.56	
	Beam (k=4)	0	0	**0**	**0.93**	
Llama-3.1-8B / DocRED (50)	Greedy	0	0	1.58	6.85	
	Beam (k=4)	0	0	**0**	**0.94**	
Mistral-7B / SciERC (551)	Greedy	0	0	13.40	1.27	
	Beam (k=4)	0	0	13.79	**0.57**	
Mistral-7B / DocRED (50)	Greedy	0	0	28.37	0.34	
	Beam (k=4)	0	0	**21.37**	**0.19**	

E Sampling Robustness

F Beam Search Compatibility

OntoCoDec's constraint enforcement is designed to be decoding-strategy agnostic. The main experiments (§5) use greedy decoding; Appendix E validates nucleus sampling. Here we verify compatibility with beam search, completing coverage of the three standard decoding strategies.

Implementation challenges. Two stateful components require adaptation for beam search. (1) The grammar layer (XGrammar) maintains an internal `GrammarMatcher` state that tracks the generated token sequence; beam reordering invalidates this state because beams are permuted across steps. Our solution instantiates a fresh `GrammarMatcher` per beam per step and replays the beam's current token sequence through XGrammar's stateless accept API (64 lines of code). (2) The semantic layer's `TypeConstraintProcessor` accumulates entity and relation registries that assume a single generation path. Our solution isolates state per beam by calling `detect()` on the full partial output at each step, reconstructing all entity/relation state from scratch for each beam independently (61 lines of code). Both adaptations add ∼125 lines total without modifying the existing greedy/sampling logic.

Results. Table 15 compares beam search (k=4, `early_stopping=True`) against greedy decoding under full S1 constraints across six model-dataset combinations.

Discussion. For calibrated models (Qwen3-8B, Llama-3.1-8B), beam search achieves DR=0% across all four model-dataset combinations (both SciERC with 7 entity types, 551 documents, and DocRED with 96 relation types, 50 documents), regardless of schema complexity. On SciERC, beam search eliminates DR for both Qwen (0.93% → 0%) and Llama (0.08% → 0%), and substantially reduces CF (Qwen: 9.90% → 0.53%; Llama: 6.56% → 0.93%). On DocRED (matched 50-document subsets), beam search eliminates DR for both Qwen (8.69% → 0%) and Llama (1.58% → 0%), with CF reductions of 74% and 86% respectively. Llama DocRED shows the largest R-F1 gain (+3.4 pp), consistent with beam search recovering valid type assignments that greedy decoding missed due to constraint fallback. The CF reduction is an indirect effect: beam search reduces constraint-fallback frequency by exploring alternative token sequences that satisfy both grammar and semantic masks simultaneously, avoiding the fallback-to-grammar path that produces cross-field inconsistencies under greedy decoding.

Mistral-7B-v0.3 constitutes a boundary condition on both datasets. On SciERC (7 entity types), beam search does not reduce DR (13.4% → 13.8%), though CF drops by 55% relative (1.27% → 0.57%) and R-F1 improves by +0.7 pp; the unchanged DR confirms that all residual violations stem from entity type misclassification, not constraint-search failure. On DocRED (96 relation types), beam search reduces DR by 25% relative (28.4% → 21.4%): the more complex schema introduces constraint-search fail-

ures that beam exploration resolves, but the entity misclassification component remains. This contrast reveals that beam search resolves constraint-search failure (where greedy decoding encounters $G(t) \cap S(t) = \emptyset$ conflicts) but cannot overcome fundamental model calibration limitations in entity typing.

Beam width ablation. Table 16 varies beam width $k \in \{1, 2, 4, 8\}$ on Qwen3-8B / SciERC (551 documents) under full S1 constraints.

Table 16: Effect of beam width on constraint satisfaction and extraction quality (Qwen3-8B, SciERC, 551 documents, full S1 constraints). k=1 corresponds to greedy decoding.

k	DR↓	CF↓	E-F1↑	R-F1↑
1 (greedy)	0.93	9.90	30.98	8.36
2	0.35	4.12	31.54	**8.75**
4	**0**	0.53	32.00	8.56
8	**0**	**0.08**	**32.18**	8.39

DR elimination at $k \geq 4$ confirms that constraint-search failure is resolved with moderate exploration width. CF continues to decrease with larger beam (0.08% at k=8, only 2 residual violations), while entity extraction quality improves monotonically. k=4 represents the optimal cost-effectiveness trade-off: it eliminates all DR violations at roughly half the compute cost of k=8.

Composability with fine-tuning. Under greedy decoding, the fine-tuned Qwen3-8B (LoRA, SciERC) with full S1 constraints already achieves DR=0% but retains CF=1.18% (21 cross-field violations across 551 documents), with E-F1=67.05 and R-F1=43.77. Adding beam search (k=4) eliminates all residual CF violations (1.18% → 0%), yielding zero violations across all four metrics (ET=0%, RV=0%, DR=0%, CF=0%) while preserving extraction quality (E-F1=66.88, R-F1=44.23; changes within ±0.5 pp). Llama-3.1-8B (LoRA, SciERC) with full S1 constraints and beam search (k=4) likewise achieves zero violations across all four metrics (ET=0%, RV=0%, DR=0%, CF=0%) with R-F1=46.11; the zero-violation result thus generalizes across calibrated model families. The three constraint mechanisms (grammar masking, semantic domain-range filtering, and beam exploration) are thus fully composable: each addresses a distinct violation source without interfering with the others or degrading task performance.

Beam search on DocRED introduces a modest increase in parse failures (3/50, 6%) compared to greedy (1.1%), as longer beam-generated sequences occasionally produce malformed JSON in complex documents. With DR=0 in 50 DocRED documents for both calibrated models, the Clopper-Pearson 95% CI upper bound is 5.8%, so the zero-DR result is statistically meaningful despite the limited sample. Combined with the sampling robustness results (Appendix E), these findings confirm that OntoCoDec's constraint enforcement is compatible with greedy, sampling, and beam search decoding without modification to the core constraint logic.

Latency overhead. Table 17 reports wall-clock latency for greedy and beam search (k=4) under LoRA fine-tuning.

Condition	**s/doc**	**Overhead**
Greedy unconstrained	2.612	1.00×
Greedy constrained (S1)	2.873	1.10×
Beam-4 unconstrained	3.446	1.32×
Beam-4 constrained (S1)	4.013	1.54×

Table 17: Latency per document (SciERC, 551 docs, Qwen3-8B LoRA FT, RTX PRO 6000 Blackwell). Overhead relative to greedy unconstrained.

Beam search itself adds 32% latency (4 beams); constraint overhead increases from 10% (greedy) to 16.5% (beam) because each beam independently maintains grammar and semantic state. The total 1.54× overhead is acceptable for offline batch IE; `early_stopping` limits additional compute to partial-sequence exploration.

We also explored a type-consensus constraint that forces identical entity types across all relations within a document. However, CF violations occur precisely where the consensus type conflicts with domain-range constraints, creating a fundamental CF↔DR trade-off. With DR-aware enforcement (skipping consensus when it would violate domain-range), the layer becomes a no-op (100% skip rate on 551 SciERC documents and 50 DocRED documents). This confirms that beam search resolves the CF↔DR tension through a fundamentally different mechanism: token-level exploration of paths satisfying both constraints simultaneously.

G Reproducibility Details

Fine-tuning configuration. All four fine-tuned models (Qwen3-8B and Llama-3.1-8B-Instruct on

SciERC and DocRED) use identical LoRA hyperparameters: rank r=16, α=32, dropout 0.05, no bias, targeting all attention and MLP projection layers (`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`). Training uses AdamW with learning rate 2×10^{-4}, cosine schedule, warmup ratio 0.1, effective batch size 16, bf16 precision, gradient checkpointing, and completion-only loss for 3 epochs. SciERC trains on 1,762 examples (260 dev) for 333 steps; DocRED trains on 3,053 examples (998 dev) for 573 steps. Maximum sequence length is 2,048 (SciERC) and 4,096 (DocRED). All runs use seed 42.

Hardware and software. Training runs on a single NVIDIA RTX PRO 6000 Blackwell Server Edition GPU (98 GB VRAM). SciERC fine-tuning takes approximately 11 minutes per model; DocRED takes approximately 1.5 hours. Software versions: PyTorch 2.10, Transformers 4.57, PEFT 0.19, TRL 1.1, Accelerate 1.13.

Inference time. Inference timing is reported in Appendix C (NVIDIA RTX PRO 6000 Blackwell Server Edition GPU, same hardware as training). Full S1 constrained decoding adds +13.1% wall-clock overhead on SciERC and +2.43% on DocRED (Qwen3-8B, zero-shot greedy); under LoRA fine-tuning, overhead is +10.0% on SciERC.

H Effect of Document Length on Constraint Effectiveness

To examine whether OntoCoDec's constraint effectiveness varies with document complexity, we partition test documents into three length bins (tertile split by word count) and report per-bin violation counts and F1.

Table 18: Document length breakdown on SciERC (Qwen3-8B). Tertile boundaries: Short $\leq$ 19 words, Medium $\leq$ 27, Long $>$ 27.

Bin	Mode	ET	RV	DR	CF	Total	E-F1	R-F1
Short (196)	Uncon.	1	5	57	0	63	32.4	8.9
	Const.	0	0	5	5	10	32.4	8.9
Medium (180)	Uncon.	6	9	104	0	119	27.8	6.7
	Const.	0	0	7	15	22	27.8	6.7
Long (175)	Uncon.	3	10	104	1	118	29.9	7.0
	Const.	0	0	4	20	24	30.0	7.3

The violation reduction rate is consistent across all length bins: 80–84% for SciERC and 74–75% for DocRED. Longer documents produce more violations in the unconstrained setting (e.g., DocRED long-bin: 3,550 vs. short-bin: 2,445), yet OntoCoDec achieves proportionally similar reduction; the constraint framework scales reliably with document complexity. ET and RV violations are fully eliminated across all bins. F1 scores are preserved or slightly improved under constrained decoding, with DocRED showing +0.9–1.5 E-F1 and +0.2–0.6 R-F1 improvement in the constrained setting.

Table 19: Document length breakdown on DocRED (Qwen3-8B). Tertile boundaries: Short $\leq$ 160 words, Medium $\leq$ 208, Long $>$ 208.

Bin	Mode	ET	RV	DR	CF	Total	E-F1	R-
Short (336)	Uncon.	177	1238	1021	9	2445	61.1	
	Const.	0	0	495	138	633	62.6	
Medium (332)	Uncon.	147	1742	1047	5	2941	59.3	
	Const.	0	5	544	180	729	60.7	
Long (330)	Uncon.	176	2209	1162	3	3550	53.9	
	Const.	0	0	731	200	931	54.8	

Table 20: Violation reduction percentage by document length bin (Qwen3-8B, full S1 constraints).

	SciERC		DocRED	
Bin	Viols (U→C)	Red.%	Viols (U→C)	Red.%
Short	63 → 10	84.1	2445 → 633	74.1
Medium	119 → 22	81.5	2941 → 729	75.2
Long	118 → 24	79.7	3550 → 931	73.8

I Prompt Sensitivity Analysis

To verify that OntoCoDec's constraint enforcement is robust to prompt variation, we evaluate three prompt variants on SciERC using Qwen3-8B (551 test documents): **Default** (the prompt used in all main experiments), **Minimal** (a shortened version listing only entity/relation types), and **Detailed** (an extended version with type definitions and extraction guidelines). All three variants share the same output JSON schema; only the instruction text differs. Each variant is decoded in both unconstrained and constrained (full Stage 1) modes.

Table 21 shows that constrained decoding eliminates entity-type and relation-vocabulary violations entirely (ET and RV drop to 0%) for all three prompt variants. Domain-range violations are reduced by 94.0%–94.6%, with a standard deviation of only 0.3 percentage points across prompts. Meanwhile, F1 scores remain effectively

Table 21: Prompt sensitivity results on SciERC (Qwen3-8B, 551 documents). ET, RV, DR denote entity-type, relation-vocabulary, and domain-range violation rates (%), respectively. ΔDR shows the relative reduction from unconstrained to constrained decoding.

Prompt	Mode	ET%	RV%	DR%	ΔDR	E-F1	R-F1
Default	Unconstrained	0.40	1.39	15.37	94.0%	31.01	8.22
	Constrained	0.00	0.00	0.93		30.98	8.36
Minimal	Unconstrained	0.58	1.16	16.91	94.6%	31.98	7.91
	Constrained	0.00	0.00	0.92		31.99	7.85
Detailed	Unconstrained	0.22	0.45	11.07	94.2%	33.55	10.48
	Constrained	0.00	0.00	0.64		33.56	10.48

Table 22: OntoCoDec corrects three Feature-of DR violations. Red = violation; green = corrected.

Triple	Head	Tail	DR
Unconstrained (3 violations)			
Feature-of(recover, shapes)	Task	OtherSciTerm	✗
Feature-of(recover, poses)	Task	OtherSciTerm	✗
Feature-of(recover, transp. obj.)	Task	OtherSciTerm	✗
Used-for(model ap-proach, recover)	Method	Task	✓
OntoCoDec (0 violations)			
Feature-of(recover, shapes)	Task	Task	✓
Feature-of(recover, poses)	Task	Task	✓
Used-for(model ap-proach, recover)	Method	Task	✓

unchanged: the largest shift is 0.13 percentage points (R-F1, Default prompt), so constraint enforcement does not degrade extraction quality.

Notably, prompt variation has a measurable effect on baseline (unconstrained) violation rates, with DR ranging from 11.1% (Detailed) to 16.9% (Minimal). Under constrained decoding, this range narrows to 0.64%–0.93%, a 20× reduction in cross-prompt variance. This convergence is expected: because OntoCoDec enforces constraints at the token level during decoding, it operates independently of prompt content. The consistent reduction rates across diverse prompt styles confirm that the mechanism is orthogonal to prompting and generalizes without prompt-specific tuning.

J Error Analysis and Case Studies

We present four representative cases illustrating constraint behavior across violation types. Cases 1–3 demonstrate OntoCoDec's corrective mechanisms; Case 4 illustrates a limitation.

J.1 Domain-Range Correction

Input (SciERC, Qwen3-8B): *"We present a novel model-based approach to recover the shapes and the poses of transparent objects from known motion."*

The unconstrained model assigns OTHERSCIENTIFICTERM to "shapes" and "poses," producing FEATURE-OF(TASK, OTHERSCIENTIFICTERM) triples that violate domain/range constraints. The semantic layer redirects the tail type to TASK, yielding schema-compliant triples (Table 22).

J.2 Dual-Layer Vocabulary and Type Correction

Input (DocRED, Qwen3-8B): *"Skai TV is a Greek free-to-air television network based in Piraeus. It is part of the Skai Group, one of the largest media groups in the country."*

The unconstrained model generates natural-language relation names (e.g., "country of origin") instead of Wikidata P-codes, triggering RV violations. The grammar layer forces valid P-codes; the semantic layer ensures type compatibility (Table 23).

Table 23: Layer 1 eliminates RV violations; Layer 2 corrects DR types on DocRED.

Unconstrained Output	Violation	Type
rel="country of origin"	Not in schema	RV
rel="headquarters location"	Not in schema	RV
rel="part of"	Not in schema	RV
rel="location of formation"	Not in schema	RV
OntoCoDec Output		
rel=P495 (Skai TV, Greece)	ORG→LOC	✓
rel=P159 (Skai TV, Piraeus)	ORG→LOC	✓
rel=P127 (Skai TV, Skai Group)	ORG→ORG	✓
rel=P17 (Piraeus, Greece)	LOC→LOC	✓

J.3 Scale Effect on Schema Compliance

Input (SciERC): *"This paper investigates the utility of applying standard MT evaluation methods (BLEU, NIST, WER and PER) to building classifiers to predict semantic equivalence and entailment."*

Both Qwen3-8B and Llama-3.1-70B-Instruct (AWQ INT4) mistype "MT evaluation methods" as METHOD (gold: METRIC), but the downstream constraint behavior differs (Table 24). The 70B model produces fewer violations (2 vs. 4) and

chooses a different relation (Part-of vs. Hyponym-of), yet OntoCoDec corrects both.

Table 24: 8B vs. 70B on the same input. OntoCoDec corrects violations at both scales; CF arises when type corrections conflict with entity-list assignments.

Setting	Relation	Types	DR
Qwen3-8B Unconstrained			
Hyponym-of(MT eval, BLEU)	Meth→Metric	✗	
Hyponym-of(MT eval, NIST)	Meth→Metric	✗	
Hyponym-of(MT eval, WER)	Meth→Metric	✗	
Hyponym-of(MT eval, PER)	Meth→Metric	✗	
Qwen3-8B OntoCoDec			
Hyponym-of(MT eval, BLEU)	Meth→Meth	✓	
Hyponym-of(MT eval, NIST)	Meth→Meth	✓	
CF: BLEU typed Metric in entity list, Method in relation			
Llama-3.1-70B Unconstrained			
Part-of(BLEU, MT eval)	Metric→Meth	✗	
Part-of(NIST, MT eval)	Metric→Meth	✗	
Llama-3.1-70B OntoCoDec			
Part-of(BLEU, MT eval)	Metric→Generic	✓	
Part-of(NIST, MT eval)	Metric→Generic	✓	

J.4 Limitation: Trajectory Divergence under Constraints

Input (SciERC, Mistral-7B): *"Amorph recognizes NE items in two stages: dictionary lookup and rule application."*

The gold annotation types "Amorph" as METHOD and "NE items" as OTHERSCIENTIFICTERM, with Used-for(Amorph, NE items). The unconstrained model mistypes "Amorph" but coincidentally produces the correct relation. Under OntoCoDec, modified token probabilities alter the generation trajectory entirely and produce a different (non-gold) relation (Table 25). This illustrates a fundamental property of autoregressive generation under modified distributions: constraint enforcement can change the entire output path, not just individual token choices.

Table 25: Constrained decoding changes the generation trajectory. The unconstrained model coincidentally produced the gold triple despite type errors.

Setting	Output	Gold?
Gold	Used-for($\text{Amorph}^{\text{Meth}}$, $\text{NE items}^{\text{OtherSci}}$)	–
Unconstr.	Used-for($\text{Amorph}^{\text{OtherSci}}$, $\text{NE items}^{\text{Task}}$)	✓
OntoCoDec	Used-for($\text{Amorph}^{\text{OtherSci}}$, $\text{recognizes}^{\text{Gen}}$) Feature-of($\text{NE items}^{\text{Task}}$, $\text{in two stages}^{\text{Gen}}$)	✗ ✗

Scale	Types	Rels	Compile (s)	Overhead
Small (SciERC)	6	7	0.33	1.02×
Medium (DocRED)	6	96	0.32	1.03×
Large (synth.)	100	500	0.35	1.03×
XLarge (synth.)	200	1,000	0.42	1.03×

Table 26: Compilation time (one-time) and per-token inference overhead across ontology scales. Small and Medium use real ontologies; Large and XLarge use synthetic ontologies with random domain/range constraints. Overhead measured on Qwen3-8B, 10 documents per scale.

K Ontology Scalability

Grammar compilation scales sub-linearly (0.33 s to 0.37 s for 7→1,000 relations); semantic constraint compilation grows linearly but remains negligible (0.4 ms to 17 ms). Inference overhead is constant at 2–3% because XGrammar uses precompiled bitmasks ($O(1)$ per token) and the semantic layer checks only the active relation's domain/range set rather than the full ontology. The practical limit is first-token uniqueness: when entity types exceed ∼200, prefix collisions in the tokenizer vocabulary may require fallback to multi-token disambiguation (§3.3).

L Full Fine-Tuning Baseline

To verify that constraint effectiveness generalizes beyond LoRA, we evaluate Qwen3-8B with full-parameter fine-tuning on SciERC (same data, 3 epochs, lr=2×10^{-4}, seed 42).

	E-F1	R-F1	DR%	CF%
Full FT unconstrained	64.78	37.89	2.74	0
Full FT + S1	64.78	37.89	**0**	1.31
LoRA unconstrained	66.86	43.65	0.31	0
LoRA + S1	67.05	43.77	**0**	1.18

Table 27: Full fine-tuning vs. LoRA on SciERC (Qwen3-8B, 551 docs). Constraints eliminate DR under both methods. Full FT underperforms LoRA, likely due to an untuned learning rate (2×10^{-4}, standard for LoRA but high for full FT). F1 is unaffected by constraints in both cases.

Full fine-tuning retains 2.74% DR violations (30 triples), concentrated in Part-of (14.6%), Hyponym-of (9.6%), and Evaluate-for (8.4%). S1 eliminates all 30 violations with no F1 cost, which confirms that constraint enforcement is effective regardless of the fine-tuning method. The CF↔DR trade-off persists (CF 0%→1.31%), consistent with the LoRA pattern (1.18%).